

# **IT Social and professional problems**

## **Lecture**

**Arkadiusz Liber**

# Lecture topics 1

- **Introduction. Basic concepts**
- **Information technology product. Preparation, design, production and exploitation of software in social and legal context.**
- **Ownership of an information technology product. Intellectual property. Definitions, practical and legal meaning, examples**
- **IT product as a subject of copyright. Subject and object of copyright. Specific legal regulations concerning the objects and subjects related to information technology.**
- **Creation and use of a computer product as a subject of copyright. Authorship of individual and collective works. Personal copyrights, manner of protection and scope of use.**

# Lecture topics 2

- **Personal rights of the author of a computer product. Personal copyright. Attributes of personal copyright and the scope of their protection.**
- **Property rights in a computer product. Property copyright and its application. Examples of protection of an information technology product.**
- **Permitted use of computer product and other copyrightable subject matter. Public use. Exclusions from protection.**
- **Copyright in scientific works and works of an educational nature. Copyright In scientific and educational institutions.**
- **Producing a software product while respecting copyright law. Creation of software and documentation with respect to copyright.**

# Lecture topics 3

- Producing a software product while respecting copyright law. Creation of software and documentation with respect to copyright.
- Criminal and professional liability related to related to computer products. Computer crimes. Forensic research.
- Information technology product and industrial property. Law of industrial property protection. Definitions. Scope of application.
- Patents and trademarks and their use in developing information technology products. Registration. Regulations concerning protection of industrial property. Regulations in Poland and Europe. Industrial property law and copyright in ethical and social context.

# Lecture topics 4

- **Licenses**
- **Ethical issues and risks in the information technology profession. Collective copyright management. Reliability and legal security of software. Codes of ethics in informatics profession.**

# Lecture - Thematic Sequence

Introduction

IT product

Civil and intellectual  
property  
and intellectual property  
in computer science

PI - as a subject of  
copyright

PI - as a subject of  
copyright cont.

PI Personal Rights

Property Rights  
Relating to PI

Permitted use of PI  
and parts thereof

PI in science  
and education

Production of PI

Criminal and  
professional liability

PI and industrial  
property

Patents and  
trademarks

Licenses

Ethics. Risks. Codes



# Organizational matters

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- Rules of participation in classes
- Activity during the class
- Credit Policy. Active participation. Quizzes. Colloquium (deadline for colloquium 9th week – **13.12.2021**)
- Contact with the instructor
- Consultation dates
- Literature



# Introduction

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# Lecture - Thematic Sequence

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- **The information technology profession and its environment**
- **Academic and professional titles in computer science**
- **Scientific disciplines in computer science**
- **The essence of computer science hardware and software**
- **Precise definitions in computer science vs. legal context**

# Science, ethics, academic and professional titles

- Science - an independent part of culture, a process of systematic study and research.
- Ethics - the branch of philosophy that deals with the study of morality.
- Occupational title - a certificate of possession of specific knowledge and skills for a particular occupation.
- Academic title - is associated with specific scientific achievements, granted according to the provisions of the law.
- Computer Science - Computer Science.

# Informatics

- As "Computer Science" or the science that deals with computers includes both hardware and software.
- Computer science is a very broad science. As such, it incorporates many disciplines of knowledge.
- Disciplines of knowledge narrow the subject matter and allow for a deeper understanding of the phenomena and detailed problems in their scope.

# Disciplines of knowledge in computer science

- Theoretical Computer Science.
- Theory of Computation.
- Computer Architecture.
- Software Engineering.
- Algorithms and Data Structures.
- Information and Coding Theory.
- Formal Methods.
- Programming Language Theory.
- Artificial Intelligence.

# Disciplines of knowledge in computer science (continued)

- Computer Graphics.
- Computer Networks.
- Databases.
- Concurrent, Parallel and Distributed Systems.  
and distributed systems.
- Real Time Systems.
- Computer Security.

# Hardware

- Computers can be based on a variety of physical phenomena.
- Computers can come in a variety of forms.
- Known technical developments in computer design:
  - semiconductors,
  - Mechanical,
  - electromechanical,
  - Optical,
  - analog,
  - Digital,
  - quantum,
  - biological.



# Computer hardware and its programming

- In theory, computer hardware is most often described as machines that have memory states that can be changed by a sequence of symbols that make up a program and input data. In computers with practical applications, there is the ability to generate output symbols that are the results of calculations.
- The ways in which computers are programmed differ significantly because of the difference in the physical phenomena used and with the ability to affect the change of states using these phenomena.
- For example, in semiconductor computers, states are stored in semiconductor memory cells (e.g., charges stored in the capacitance which is accessed using a transistor key). Using the two charge values, a binary machine is created on which functional blocks (registers, adders, counters, etc.) are implemented secondarily. Programming here is based on the sequential change of binary states of the memory.

## Computer hardware and its programming cont.

- In quantum computers, programming looks a little different.
- Taking as an example a quantum computer realized on the basis of NMR spectrometer in one moment all molecules placed inside the electromagnet are being processed. The processing here is always parallel and the memory states are represented here by the magnetic moments of the molecules.
- Programming here is not about changing the magnetic moments of specific molecules individually, but about sequentially changing all the magnetic moments of the molecules in the cavity of the apparatus so as to produce a specific end result. The programming here takes the form of a superposition of quantum unitarity operators.

# Software-related terms

- Algorithms and heuristics.
- Information.
- Data.
- Computer Program.
- Database.
- Data Structure.
- Programming language.

It is important to keep in mind that definitions of particular terms may differ from one another in different literature sources. Most often the differences are due to historical reasons, mathematical clarification, or context used.

# Algorithm. Examples of axiomatic definitions according to Burgin, 1985

## Algorithm

- Each algorithm determines a binary relation  $R$  on the Cartesian product  $X \times Y$ , where  $X$  the set of inputs of the algorithm,  $Y$  the set of outputs of the algorithm.

## Deterministic algorithm

- Any partial function  $f_A$  that maps a set of inputs  $X$  into a set of outputs  $Y$ .
- A deterministic algorithm is obtained from the axiom of algorithm by imposing constraints on the relation  $R$ . It is required that the relation be a partial function.
- The concept of partial function is a generalization of the concept of function. A partial function is a mapping  $f: X \rightarrow Y$  such that there exist subsets  $X'$  of the set  $X$  such that  $X' \rightarrow Y$  is a function according to the traditional meaning. Partial functions are used, among other things, in algorithm theory in cases where the set  $X$  cannot be defined exactly.

## Important legal context

- Precise knowledge of basic concepts is extremely important, for example, in the forensic examination of hardware and software.
- The result of a forensic examination is relied upon by parties during civil and criminal proceedings.
- For example, in a copyright infringement action against a computer program, it is fundamental to determine whether the designated work is actually a computer program, who is the subject and what is the actual subject matter of the copyright. Further to the facts of the case.

# IT product

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# Lecture - Thematic Sequence

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# Issues

- **Creation and features of an IT product including software**
- **IT product in social and legal context**
- **Social and professional problems arising in connection with the production and use of IT products**

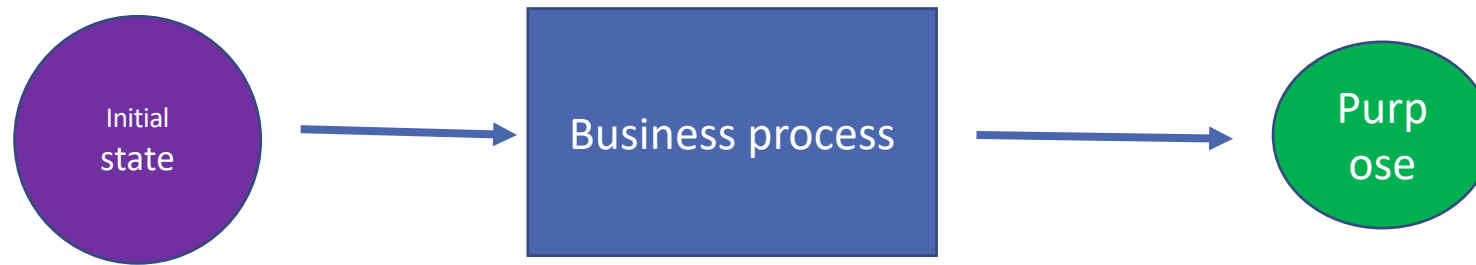


## IT product and its realization in the social and legal context

- Product - anything that can be offered to the market (marketing-economic meaning of product).
- IT product - a product that incorporates IT solutions.
- Examples of IT products:
  - computer hardware,
  - SOFTWARE,
  - software projects,
  - results of IT audits and analysis,
  - A significant proportion of modern industrial products (e.g. cell phones, modems, computerized household goods,
  - etc.

# The nature of IT product development

- The process of developing an IT product is an example of a business process.
- A business process is a sequence of interrelated activities leading to a specific outcome.



- There is a growing share of software in today's IT products.
- New opportunities are emerging for product personalization and software control of product performance.
- It is becoming possible to produce better and better products with software locking functionality and timing.

## **New opportunities to influence the product lifecycle related to related to IT products**

- In the case of a classic product such as a sheet of steel, for example, the material must be modified by doping and processing in order to influence its life cycle. Physical and chemical processes affect hardness, corrosion rates and elasticity.
- If we wanted to produce sheets that could only be used for 3 years, we would have to spend many years researching and experimenting on appropriate doping and production methods. Once the sheets are sold, of course, we would have no way of remotely extending or shortening their lifespan.

## **New opportunities to influence the product lifecycle related to IT products**

- The situation is different in the case of an IT product. In such a product there is software that affects the operation of the product.
- Thus, it is extremely easy to influence the life cycle of a product programmatically.
- What is more, through update mechanisms, the rules of operation of a device can be changed during its use. For example, shorten its correct operation or limit software functionality over time.

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# Ownership of an IT product

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# Issues

- **Ownership in the civil law sense**
- **Intellectual property**
- **Specific features of intellectual property**

# Property

- Ownership is a term with a legal meaning.
- Contemporary definitions of property are based on:
  - A positive approach to the definition of property,
  - A negative approach to defining ownership,
  - mixed and other legal system-specific approaches.
- The definition of property in the positive sense consists in the enumeration by the legislator of the powers of the owner (in addition to the enumerated powers, no other powers are presumed).
- The definition of property in the negative sense is that the owner can dispose of his property in any way he wishes outside of the explicitly enumerated restrictions.

# Ownership in Polish law

- Ownership - A property right that allows to **use and disposal of** a thing to the **exclusion of persons other** than the owner.
- Certain property rights:
  - **possession of** things,
  - **Disposition of** the thing (including destruction of the thing),
  - **USE**,
  - **taking benefits** from things (e.g., leasing, renting, borrowing, natural benefits from crops),
  - **Disposition of** property (transfer of property, donation, relinquishment of property, disposition on death).

# Limitations of ownership

The use and disposal of a thing **must be lawful**.

A change in the acts of law may materially alter the scope of use and disposal of the thing.

# Co-ownership

- Joint Ownership - The right of ownership of a thing vested indivisibly in more than one person.
- Each co-owner is entitled to all the attributes of property rights.
- In Polish law, co-ownership can be:
  - joint, such as a community of property,
  - in fractional parts.
- Joint ownership is not the same as commonality.

## Presumption of equal distribution

Unless otherwise agreed it shall be presumed that in the case of joint ownership in fractional parts, the shares are equal.

This presumption is rebuttable. This means that it is possible to change the amount of shares, for instance through the courts.

# Common property management

- The management of the common thing follows:
  - under the terms of the co-owners' agreement,
  - under the terms of the Civil Code,
  - based on a court order.
- Board Actions:
  - ordinary,
  - exceeding ordinary management,
  - conservative.

# Abolition of joint ownership

- Any co-owner may request that the co-ownership be abolished.
- The right to demand abolition of co-ownership can be limited by an agreement between the owners, but for no longer than 5 years.
- The abolition of co-ownership shall take place by way of:
  - an agreement between co-owners (in some cases a notarial deed is required),
  - court order.
- There may be limitations, for example, in the case of a farm.



# Intellectual property

- A group of legal monopolies that includes intangible property as defined by **copyright** and **patents, industrial designs, and trademarks**.
- Patents and trademarks are so-called industrial property.
- The term intellectual property was first used in 1888.
- In 1967, **the World Intellectual Property Organization**, WIPO, was created.

# **Computer product as a subject of copyright**

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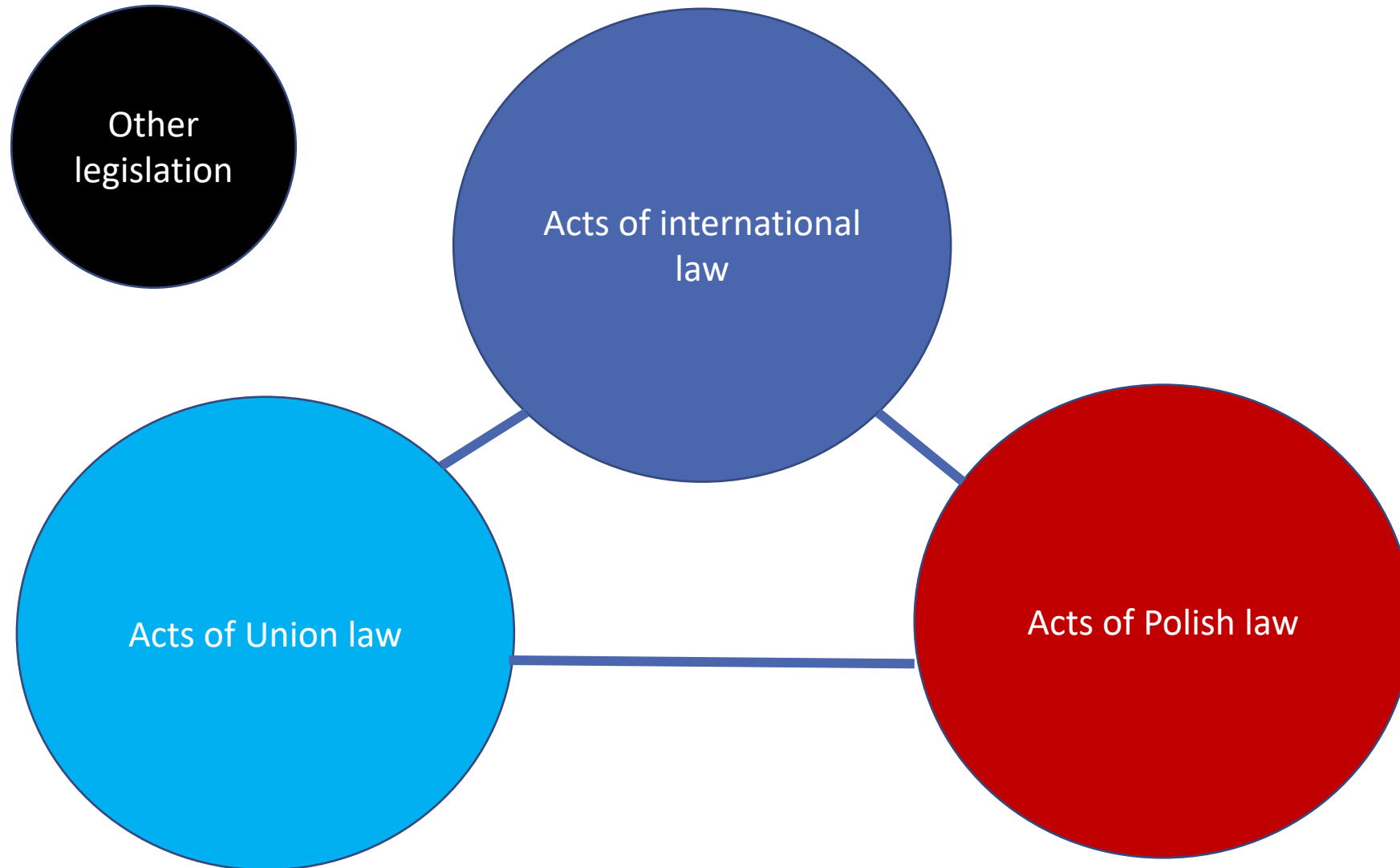
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- **IT product and its protection**
- **Subject and object of copyright**
- **Specific objects of copyright**
- **Examples of copyright subjects in computer science**
- **What can't be copyrighted**
- **Dualism in Polish Copyright Law**

# Sources of law



## **Acts of international law**

- International Convention for the Protection of Performers...(1961).
- Paris Act of the Berne Convention...(1971).
- World Intellectual Property Organization Treaty...(1996).
- Agreement on commercial aspects of intellectual property rights...(1996).

## Acts of Union law

- European Parliament and Council Directive:
  - No. 96/9/EC...(1996),
  - No. 2000/31/EC...(2000),
  - No. 2001/29/EC...(2001),
  - No. 2001/84/EC...(2001),
  - No. 2004/48/EC...(2004),
  - No. 2006/115/EC...(2006),
  - No. 2006/116//EC...(2006),
  - No. 2009/24/EC...(2009).
- Council Directive 93/83/EEC...(1993).

# Acts of Polish law

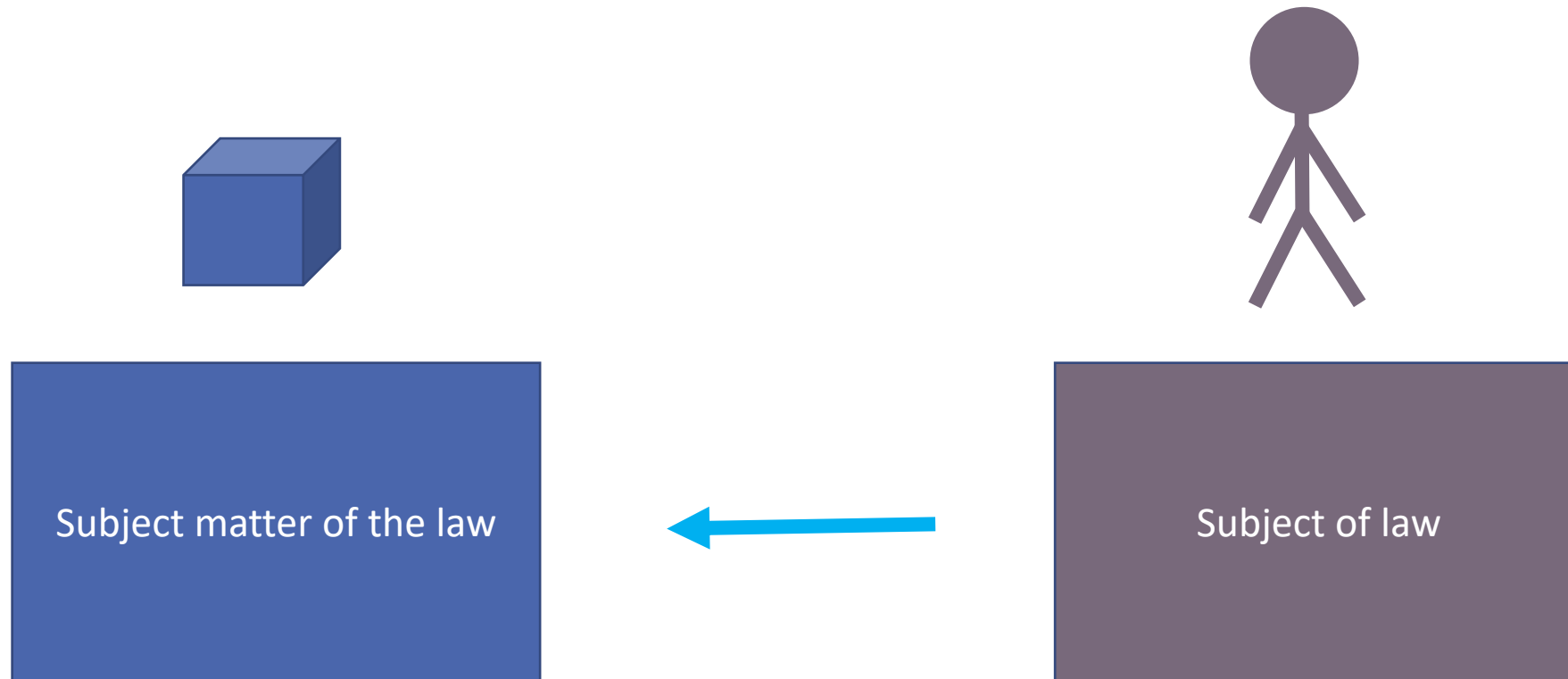
- Copyright and Related Rights Act...(1994-).
- Higher Education Act...(2005-).
- Law: Civil Code...(1964-).
- Law: Criminal Code...(1997-).
- Electronic Services Act...(2002-).
- Database Protection Act...(2001-).
- Act on the Polish Academy of Sciences...(2010-).



# Acts of Polish law

- Regulations of the Minister of Culture:
  - 17.01.2003,
  - 24.02.2003,
  - 2.06.2003,
  - 27.06.2003,
  - 30.04.2004.
- Regulations of the Minister of Science and Higher Education:
  - 14.03.2007,
  - 6.12.2006.

# Dualism



## Subject matter of copyright

- Subject matter of copyright - any manifestation of creative activity of **individual character**, determined **in any form**, regardless of value, purpose and manner of expression (work)  
value, purpose and manner of expression (work).

# Specific objects of copyright

- expressed in words, mathematical symbols, graphic signs (literary, journalistic, scientific, cartographic and computer programs);
- Plastic;
- PHOTO;
- lutherie;
- industrial design;
- Architectural, architectural and urban planning;
- Musical and Word and Music;
- stage, stage-music, choreography and pantomime;
- Audiovisual (including film).

## Comments on the list of specific copyright subjects

- Despite the fact that the Polish legislation lists the specific objects of copyright literally, it should be remembered that not always the object of the right listed is a subject of copyright.
- To be the subject of copyright, the premise of its individual creative character must be met.
- Individual creative character does not mean that the work was made by an individual person, but that it has individual creative qualities that other works do not have.

# Software example

- Let's take an example code:

```
void main()  
{  
}
```

- It is undoubtedly a computer program (a sequence of instructions in some popular programming language).
- As a computer program, it is therefore included in the list of specific copyright subjects.
- However, the piece of software above is not copyrightable. It does not have an individual creative character and has a typical, template form, in addition such that it must appear in every program written in this programming language.

# Copyrightable subject matter arising from another's work

- Translation.
- Conversion.
- Adaptation.
- Another study ?
- The disposition and use of the development is subject to the permission of the creator of the original work.
- Copies of the development must list the author and title of the original work.

## Are not the subject of copyright

- normative acts or official drafts thereof;
- official documents, materials, signs and symbols;
- published patent or protective descriptions;
- simple press releases.



# Only the manner of expression is protected

- Only the manner of expression can be protected; discoveries, ideas, procedures, methods and principles of operation and mathematical concepts are not protected.
- A work is subject to copyright from the moment it is determined, even if it is in an unfinished form.
- Protection is granted to the creator regardless of any formalities.

## Special case. Algorithms implemented in computer programs

- An algorithm implemented in a computer program should be classified as a procedure, method, operating principle and sometimes as a mathematical concept.
- As such, it is not subject to copyright.
- The consequence of this fact is that it cannot be protected under copyright law in Poland.

## Special case. Program fragments and non-functioning programs

- For a work to be subject to copyright, it must be individually creative.
- If a computer program has **an individual** creative character, the fact that it **contains errors**, does not compile or does not work does not prevent it from being considered **copyrightable subject matter** and protected under the law.
- Of course, the fact that the program **is not finished** but that parts of it have an individual creative character **does not prevent its** protection.

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## **Lecture**

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# **Creation and use of a computer product as copyrightable subject matter**

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# Issues

- **Individual and group IT products**
- **Division of rights to a multi-author copyrighted subject matter**
- **Rights and obligations of co-authors**

# Production of an IT product

- An IT product consists of the IT part, which can be up to 100% of the total product.
- If the IT part includes software then its development is subject to the well-known processes that make up the software life cycle.
- Taking a simple model of the software life cycle, we can distinguish six basic stages:
  - pre-requirements analysis,
  - design,
  - implementation,
  - testing,
  - implementation,
  - maintenance.



# Legal and ethical issues that arise prior to software development

- Software may be developed individually or in teams.  
In case of individual realization, the creator has full copyright to the part he/she created.
- In case the software is developed in a team, the creative shares of individual authors should be considered. Unless an appropriate agreement is written, all authors will by default be entitled to equal shares.
- If the team members are employed by one employer and the programming project is carried out within the employment relationship, economic copyright shall pass to the employer. However, the employer should remember about moral copyrights, including in particular the right to mark the work.
- In the case of a team in which some of the authors work under an employment relationship and some individually, by default the economic copyrights will then belong to the employer and the individual authors. In order to hold all the economic copyrights the employer should enter into agreements with the individual authors transferring the economic copyrights to the employer. To be legally effective the agreement should be in writing.

## Legal and ethical issues at the software pre-requirements analysis stage

- The analysis of prerequisites is often related to the collection of source materials. It is important to note that you should only collect copyrighted materials if you are allowed to do so (for example, if you obtain permission from the holder of the copyright to the works you are collecting). In the case of scientific or educational purposes, it should be remembered that the author still retains the right to remuneration for the use of his/her works.
- Concepts, operating principles or algorithms may be developed as part of the prerequisite analysis. It should be remembered that they are not subjects of copyright law and are not protected under Polish and European Union law.

## Legal and ethical issues at the project stage

- In the case of a project that is a concept of operation or implementation of software, it should be remembered that as a concept it cannot be protected under copyright law.
- Only the individual expression of that design (e.g., creative writing or drawing) will be protected here.
- In order to protect ideas and concepts, it is a good idea to consider making the project a business secret.
- As employees working on concepts, policies, ideas, or algorithms, it makes sense to hire quality professionals who are guided by professional ethics in their work.

## Legal and ethical issues at the implementation stage

- This step results in the source code and resultant code of the software.
- The legal situation here is relatively simple. Computer programs, as long as they have an individual creative character are subject to copyright.
- When the implementation is done using external code and libraries, the legal situation of the resulting product can be very complicated.
- External source code and libraries have their individual licenses, which very often are unacceptable for an entrepreneur. For example, they may force the entire product created using them to have an OpenSource or Freeware license.

## Legal and ethical issues in the testing phase

- Modern software testing has gone far beyond simply testing external functionality.
- Currently, the testing stage is often similar to the implementation stage. Using appropriate code snippets, software can be tested automatically.
- Implementation of testing procedures may be subject to copyright.

# Legal and ethical issues during the implementation phase

- The implementation stage should end with running the software at the customer's site and training the personnel who will be able to operate the software.
- The following legal and ethical risks are most common during the implementation phase:
  - violation of software licenses held by the customer during customization of deployed software,
  - violation of software licenses during training and trial use of the system,
  - adaptation and reverse engineering of other software to interact with software implemented in violation of license terms and copyright laws.

## Legal and ethical issues in the maintenance phase

- Software maintenance should keep the software in continuous operation.
- The main maintenance problem is aging hardware and software.
- You may need to use legacy operating systems or external programs to enable legacy components to run in the new environment to keep the software running.
- With such operations, there may be a problem with licenses.

# **Moral rights of the author of an IT product**

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- **Special features of personal copyright**
- **Author's rights**
- **Marking works with name, surname and pseudonym**
- **Author's supervision**
- **Enforcement of copyright in works of unknown authors**

# Subject of copyright

- Copyright is vested in the creator, except as otherwise provided by law.
- The author shall be **presumed to be the** person whose **name appears** as such on copies of the work or whose authorship is otherwise publicly acknowledged in connection with  
in connection with the distribution of the work.
- As long as the author has not disclosed his authorship, he is replaced in the exercise of his copyright by the **producer** or **publisher or**, in their absence, by the relevant **collecting society**.

# Anonymous programs and anonymous code snippets

- The author is not required to sign or register his works somewhere.
- This includes books, reports, programs, and source code snippets.
- An anonymous program or piece of code, as long as it has an individual creative character, is copyrightable subject matter.
- The effect of a program or code not being signed, and for example not having a publisher, is to replace it with a collecting society.

## Copyright commonality. Co-authors

- Co-authors are entitled to copyright jointly. The proportions of shares shall be presumed to be equal.
- Each co-creator may request that the size of the share be determined by the court, based on the contributions of the creative work.

# The author's right to **a part** having independent meaning

- Each co-author **may exercise copyright in his or her** independently significant **portion of the** work without prejudice to the rights of the other co-authors.

## Entitlement to the entire work

- The consent of all co-authors is needed to exercise copyright in the entire work.
- In the absence of such consent, any of the co-creators may request a decision by the court, which shall decide taking into account the interests of all co-creators.

# Claims to the Work

- Each co-author may assert a claim for infringement of copyright in the work as a whole.
- The benefit obtained shall accrue to all co-creators according to their respective shares.



## Example of a claim to a multi-author program

- The ability of each co-creator to pursue a claim for copyright infringement has far-reaching implications.
- Let's imagine that a computer program is being developed with additional contributors.
- You cannot take away the right to claim copyright in the work as a whole from any of the co-authors, such as the one whose contribution is the least.
- Of course, the financial claim obtained must be divided among all authors according to their shares. If there is no agreement on the size of the shares of the individual authors. It is due to the presumption of equality of shares that the co-author with the smallest contribution will by default receive the same fraction of the financial claim as the author with the highest creative contribution. This can be changed by an agreement between the authors or by a court decision.

## Economic Rights to the Work

- The author's economic rights vested in co-authors shall be subject to the provisions of the Civil Code on joint ownership in fractional parts.
- If there is a disparity in the parts of the work created by different authors, you should consider a contract that explicitly states the shares.

## Distribution of the combined work

- When authors have combined their separate works for the purpose of joint distribution, each author may require the other authors to grant permission to distribute the whole so formed, unless there is a just ground for refusal and the contract does not provide otherwise.

# Collective work

- The author's economic rights in a collective work, in particular in an encyclopaedia or periodical publication, shall be vested in the producer or publisher, and in the authors of the individual parts that have independent significance. The producer or publisher shall be presumed to hold the title.

# The work and the employer

- Unless otherwise stipulated by law or the employment contract, the employer whose employee has created a work as a **result of the performance of duties under the employment relationship shall**, upon acceptance of the work, acquire the author's economic rights within the limits arising from the purpose of the employment contract and the consensual intention of the parties.

## Loss of property rights by the employer

- If the employer, within a period of two years from the date of acceptance of the work, **does not proceed with the distribution of the work** designated in the employment contract for distribution, the author may establish a reasonable period of time for the employer in writing to distribute the work, with the result that, upon the unsuccessful expiration of the period, the rights acquired by the employer, together with the ownership of the object on which the work is recorded, **shall revert to the author, unless the contract provides otherwise**. The parties may set a different time limit for the commencement of distribution of the work.

## Employer, employee, and carrier of the work

- Unless otherwise specified in the employment contract, upon acceptance of the work, the employer acquires ownership of the object on which the work is fixed.
- It is important to note that the value of the object on which the work is fixed is not important. The employer acquires the object at the time of acceptance of the work made on it within the scope of the employment relationship.

# **IT Social and professional problems**

## **Lecture**

**Arkadiusz Liber**



# **Property rights to an IT product**

IT Social and professional problems

Arkadiusz Liber

# Lecture - Thematic Sequence

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- **Value of the copyrighted subject matter**
- **Economic rights in the subject matter of the copyright**
- **Transfer of economic rights to objects of copyright**
- **Special rigors in contracts for the transfer of economic rights to copyright objects**
- **Duration of the author's economic rights**

# Copyright

- Personal copyrights
- Property copyrights

# Personal copyrights

- Unless otherwise provided by law, moral rights protect the author's links, unlimited in time and not subject to waiver or sale with a work, in particular the right to:
  - 1) **authorship of** the work;
  - 2) to **mark** the work with his/her name or pseudonym or to make it available anonymously;
  - 3) **inviolability of the** content and form of the work and its fair use;
  - 4) **Decide** whether to make the work available to the public for the first time;
  - 5) **supervise the** use of the work.

# Property copyrights

- Unless otherwise provided by law, the author shall have the **exclusive right to use and dispose of the work** in all fields of exploitation and to remuneration for the use of the work.

# Fields of use

Example fields of use:

- in print form,
- on CD,
- On DVD,
- videotape recording,
- recording as mp3 file,
- network sharing,
- as a pdf file.

# Database multiplication

- The development or reproduction of a database which has the characteristics of a work, carried out by a lawful user of the database or of a copy thereof, does not require the authorisation of the author of the database if this is necessary for access to and normal use of the contents of the database. If the user is authorized to use only a part of the database, this provision applies only to that part.



# Enforcement of author's economic rights

- Author's economic rights are not enforceable as long as they serve the creator. This does not apply to maturing debts.
- After the death of the author, the heirs may oppose the execution of the copyright in an unpublished work, unless the opposition would be inconsistent with the disclosed will of the creator regarding the distribution of the work.
- The right to remuneration, is not waivable, assignable or enforceable. This does not apply to matured claims.

## Resale of original works of art and photography

- (1) The author and his/her heirs, in case of professional resale of original copies of an artistic or photographic work, shall be entitled to remuneration being the sum of the following rates:
  - 1) 5% of a portion of the sales price if that portion is included in the range up to the equivalent of €50,000, and
  - 2) 3% of a portion of the sale price if that portion is between the equivalent of 50,000.01 euros and the equivalent of 200,000 euros, and
  - 3) 1% of a portion of the sale price if that portion is between the equivalent of 200,000.01 euros and the equivalent of 350,000 euros, and
  - 4) 0.5% of a portion of the sale price if that portion is between the equivalent of 350,000.01 euros and the equivalent of 500,000 euros, and
  - 5) 0.25% of a part of the sale price, if this part is contained in the range exceeding the equivalent of 500,000 euros
- - but not higher than the equivalent of 12,500 euros.

## **Resale of original works of art and photography**

- The preceding provision shall not apply in the case of a sale price lower than the equivalent of 100 euros.
- Original copies of a work as defined here are:
  - 1) copies made personally by the creator;
  - (2) copies considered to be original copies of the work if made personally in limited quantity, by the author or under his supervision, numbered, signed, or otherwise marked by him.

# Resale of manuscripts of literary works

- The author and his/her heirs are entitled to remuneration amounting to 5% of the price of manuscripts of literary and musical works resold professionally. and musical works.

# What is resale?

- (1) A resale within the meaning of the Act is any sale that occurs after the first disposition of the copy by the creator.
- (2) Professional resale within the meaning of this Act shall be all actions of the nature of resale performed, in the course of their business, by sellers, buyers, intermediaries and other entities professionally engaged in dealing in works of art or manuscripts of literary and musical works and musical works.

## Resale of originals of other works

- The above provisions shall also apply to original copies and manuscripts of works **other than those mentioned above**, whose authors have their permanent residence in the territory of the Republic of Poland on the date of resale.

## **Fees on new and imported duplicating devices**

- Manufacturers and Importers:
- (1) tape recorders, video recorders, and other similar devices,
- (2) photocopiers, scanners, and other similar reprographic devices that make it possible to obtain copies of all or part of a copy of a published work,
- 3) clean carriers used for the fixation, within the scope of their own personal use, of works or the objects of related rights, with the use of the devices specified in items 1 and 2 - shall be obliged to pay, determined in accordance with section 5 determined pursuant to section 5, to collective management organisations acting on behalf of authors, performers, phonogram and videogram producers and publishers, fees not exceeding 3% of the amount due on sales of such devices and carriers.

## The amount of the fee on clean media

- The minister competent for culture and national heritage protection, after consultation with copyright or related rights collective management organizations, associations of authors, performers, phonogram and videogram producers and publishers, as well as organizations of producers or importers of devices and clear carriers mentioned above, shall determine, by way of regulation, the following categories of devices and carriers and the amount of fees referred to above, taking into account the capacity of a device and a carrier to reproduce works, as well as their designation to perform functions other than reproducing works, the manner of collecting and distributing the fees, as well as the organisations of collective management of copyright or related rights entitled to collect the fees.



# Distribution of amounts from the sale of tape recorders

- The amount received from fees from the sale of tape recorders and other similar devices and related blank media is attributable to:
  - 1) 50% - creators,
  - 2) 25% - performers,
  - 3) 25% - producers of phonograms.

## **Distribution of amounts from the sale of VCRs**

- The amount received from fees from the sale of VCRs and other similar devices and related blank media is attributable to:
  - 1) 35% - creators,
  - 2) 25% - performers,
  - 3) 40% - producers of videograms.

## **Distribution of amounts from the sale of reprographic devices**

- The amount received from fees from the sale of reprographic devices and associated blank media is attributable to:
  - 1) 50% - creators,
  - 2) 50% - to publishers.

## Amounts from reproduction business activities

- Holders of reprographic devices who carry out economic activity in the field of reproduction of works for the personal use of third parties, are obliged to pay, through an organisation for collective management of copyright or related rights, fees in the amount of up to 3% of receipts on that account to authors and publishers, unless the reproduction is done under an agreement with the right holder. These fees are due to authors and publishers in equal parts.

# Broadcasting of works on radio and television

- Radio and television organizations are permitted to broadcast published minor musical, spoken word and spoken-musical works **only on the basis of a contract concluded with a collecting society**, unless the right to broadcast the works commissioned by the radio or television organization is granted to it under a separate contract.

# Duration of the author's economic rights

- Subject to the exceptions provided by law, copyright expires at the end of 70 years:
- 1) from the death of the author, and for co-authored works, from the death of the co-author **who survives the others**;
- 2) with respect to a work whose author is unknown - **from the date of the first distribution**, unless the pseudonym leaves no doubt as to the identity of the author or if the author has disclosed his identity;
- 3) with regard to a work the author's economic rights to which are vested by law in  
by operation of law to a person other than the author - **from the date of distribution of the work**,  
and when the work has not been disseminated - from **the date of its determination**;

## Duration of economic copyrights cont.

- 4) with regard to an audiovisual work - from the death of the last deceased of the following persons: the main director, the author of the screenplay, the author of the dialogues, the composer of the music composed for the audiovisual work;
- 5) with respect to a verbal and musical work, if the verbal work and the musical work were created specifically for the verbal and musical work, from the death of the later deceased of the following persons: the author of the verbal work or the composer of the musical work.

# How to calculate the duration of protection

- The duration of author's economic rights shall be calculated in full years following the year in which the event giving rise to the running of the term occurred



## Fee applicable after expiration of author's economic rights

- Producers or publishers of copies of literary, musical, plastic, photographic and cartographic works that do not enjoy the protection of author's economic rights are obliged to transfer to the Fund a payment of 5% to 8% of revenues.

# Passing of copyrights

Unless otherwise provided by statute:

- 1) Copyright may pass to others by inheritance or by contract,
- 2) the purchaser of the author's economic rights may transfer them to other persons, unless the contract provides otherwise.

# **Permitted use of computer product and other copyrightable subject matter**

IT Social and professional problems

Arkadiusz Liber

# Lecture - Thematic Sequence

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- **Using copyrighted subject matter for your own purposes**
- **Permitted use**
- **Group of persons entitled to fair use**
- **Legitimate interests of the author and fair use**

# Permitted use

- It is permitted to use an already distributed work for one's **own personal use** without the author's permission. This provision does not authorize building upon another person's architectural or architectural and urban planning work, nor does it authorize the use of electronic databases that meet the characteristics of a work, except for one's own nonprofit scientific use.  
for profit.
- The scope of personal use includes the use of single copies of works by a circle of people who are **in a personal relationship,**  
**in particular by way of kinship, affinity or social relationship.**

# Incidental Reproduction of a Work

- The author's permission is not required for transient or incidental reproductions of works that have no independent economic significance but are an integral and essential part of a technological process and are intended solely to enable:
- 1) the transmission of a work in an ICT system between third parties by an intermediary or
- 2) the lawful use of the work.

## **Distribution of RTV in a group of up to 50 recipients**

- It is permissible to distribute by means of a community antenna and a cable network the works broadcast by another radio or television organization by satellite or terrestrial means, if this is done in the course of simultaneous, integral and gratuitous free of charge broadcasting of radio or television programmes and is intended for a designated audience located in a single building or in single-family houses comprising up to 50 households.



## Use of receivers in publicly accessible areas

- Owners of equipment used for the reception of radio or television broadcasts may use such equipment to receive the broadcast works, even if the equipment is located in a place that is accessible to the public, if this does not involve any financial gain.

# Dissemination for information purposes

- Free to distribute for informational purposes in print, radio and television:
- 1) already widespread:
  - (a) Reports on current events,
  - b) topical articles on political, economic or religious subjects, unless it is expressly stated that their further dissemination is prohibited,
  - (c) actual statements and reporter photographs;
- 2) brief excerpts from the reports and articles mentioned above;
- 3) reviews of publications and distributed works;

## Dissemination for informational purposes cont.

- (4) Speeches delivered at public meetings and hearings; however, this does not authorize the publication of a collection of speeches by one person;
- 5) brief summaries of distributed works. collections of one person's speech;
- The author **is entitled to remuneration** for the use of the works referred to above.

## **Citation of works presented at the event**

- Reports on current events may cite works made available at these events, but only to the extent justified by the purpose of the information.

## Use of a work under fair use

- Works may be used within the limits of fair use as long as the name of the author and the source are mentioned. The mentioning of the author and the source should take the existing possibilities into consideration. The author is not entitled to remuneration, unless otherwise provided by law.
- The permitted use **may not infringe upon the normal use of the work or harm the legitimate interests of the author.**

# **Social and professional problems**

## **Lecture**

**Arkadiusz Liber**

# **Copyright in scientific and educational works**

IT Social and professional problems

Arkadiusz Liber

# Lecture - Thematic Sequence

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- **Specific Features of a Scientific Work**
- **Textbook, presentation, and other work of an educational nature**
- **Special rules for using others' works for scientific and educational purposes**
- **Proper citation**
- **Specific rules to help you create a correct and legal thesis**
- **Plagiarism and plagiarism proceedings**

# Teaching and learning objectives

- Scientific and educational institutions may, for teaching or research purposes, use the original and translated versions of distributed works and in translation and to make copies of fragments of the distributed work for this purpose.

# Sharing of works by libraries

- Libraries, archives and schools may:
- 1) **make** copies of distributed works **available** free of charge within the scope of its statutory tasks;
- 2) **make** or have made copies of distributed works to add to, preserve, or protect their own collections;
- 3) **make collections available** for research or cognitive purposes through the **ends of the information system** (terminals) located on the premises of these units.

## Quoting excerpts from works

- It is permitted to quote in works constituting an independent whole extracts of distributed works or minor works in their entirety to the extent justified by the explanation, critical analysis, teaching or laws of the genre.
- It is permitted to include distributed minor works or excerpts from major works in textbooks and excerpts for teaching and research purposes.
- It is permitted to include distributed minor works or fragments of major works in anthologies for teaching and scientific purposes.
- In the cases mentioned above, the author is entitled to remuneration.

# Preparation of documentation studies

- Information or documentation centers may compile and distribute their own documentary works and single copies, no larger than one publishing sheet, of excerpts from published works.
- The author or a competent organization for collective management of copyright or related rights is entitled to collect remuneration from the centers mentioned above for making available, against payment, copies of fragments of works.

# Unpaid public performances

- It is permitted to perform publicly distributed works free of charge during religious ceremonies, school and university events or official state ceremonies if there is no direct or indirect financial gain involved and the performers do not receive remuneration, with the exception of advertising, promotional or electoral events.

## Displaying a copy of the work to the public

- ( 1) The owner of a copy of an artistic work may exhibit it publicly if no pecuniary gain is involved.
- (2) If it is decided to destroy an original copy of a plastic work located in a publicly accessible place, the owner shall be obliged to make an offer for sale to the creator of the work or his relatives, if communication with him to make the offer is possible. The upper limit of the price is determined by the value of the materials. If the sale is not possible, the owner is obliged to allow the author to make a copy or, depending on the type of work, appropriate documentation.

# **Producing an IT product with respect for copyright**

IT Social and professional problems

Arkadiusz Liber



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- The life cycle of an IT product
- Individual and collective production
- Buying and selling an IT product
- Special rules related to the production of an IT product in the context of an employment relationship

## **Copyright Transfer Agreement - License.**

### **Condition of specifying the fields of exploitation**

- The agreement on the transfer of author's economic rights or the agreement on the use hereinafter referred to as the "license" shall **cover the fields of exploitation expressly specified therein.**

# Invalidity of certain contractual provisions

- A contract is **invalid** in the part concerning **all** works or all works of a certain type by the same author to be created **in the future**.

## Limited to known fields of use

- The agreement can only cover fields of use that are known at the time of the agreement.
- It is therefore not possible for a publisher, without signing a new contract with the author, to distribute his works in a field of exploitation that has emerged, for example, as a result of technological progress.

## Limiting the ability to block distribution of a collective work in new fields of use

- The author of a work used or included in an audiovisual work and a work included in a collective work, after the creation of new ways of exploitation of the works, cannot, **without a valid reason, refuse to** grant permission to use that work within an audiovisual work or a collective work in the fields of exploitation unknown at the time of concluding the agreement.

## Passing on to co-authors the portion obtained by the Treasury

- If the author's economic rights of one of the co-authors were to be assigned to the State Treasury as a statutory heir, that portion shall pass to the surviving co-authors or their legal successors according to the size of their shares.

# Presumption of entitlement to author's remuneration

- If the contract does not indicate that the transfer of the author's economic rights or the granting of a license was free of charge, the author is entitled to remuneration.
- If the contract does not specify the amount of copyright remuneration, the amount of remuneration shall be determined taking into account the scope of the right granted and the benefits resulting from the use of the work.



## Remuneration vs. field of use

- Unless otherwise specified in the contract, the author is entitled to separate remuneration for the use of the work in each separate field of exploitation.

# Copyright monopoly on the exercise of subsidiary rights

- Unless otherwise specified in the contract, the author retains the exclusive right to authorize the exercise of the dependent copyright, even though the contract provides for the transfer of the entire author's economic rights.

## Author's insight into sales records

- If the remuneration of the author depends on the amount of revenue from the use of the work  
If the remuneration of the author depends on the amount of the revenues from the use of the work, the author shall have the right to be informed and to have access, to the extent necessary, to the documentation that is relevant for the determination of the remuneration.

## How to use the work if not specified

- If the contract does not specify the manner of use, it should be consistent with the nature and purpose of the work and with accepted custom.
- For example, when buying a music CD without a specific license attached, the accepted field of use is to listen to the CD from a CD player and not, for example, to rip its contents and play it back in fields of use convenient to the buyer.

## The right to change the work by a successor in title

- A successor in title, even if it acquires all the author's economic rights, **may not, without the author's consent**, make any changes to the work, unless they are necessary and the author has no legitimate reason to oppose them. This shall apply mutatis mutandis to works whose term of protection of economic copyrights has expired.

## Separate fields of exploitation

- In particular, separate fields of exploitation are:
- 1) within the scope of recording and multiplication of the work - production of copies of the work using a specified technique, including printing, reprography, magnetic recording and digital technique;
- 2) within the scope of circulation of the original or copies on which the work has been recorded - introduction into circulation, lending or rental of the original or copies;
- 3) within the scope of dissemination of the work in a manner other than specified in item 2 - public performance, exhibition, display, reproduction, as well as broadcasting and rebroadcasting
- 3) within the scope of dissemination of the work in a manner other than specified in item 2 - public performance, exhibition, display, reproduction, broadcasting and rebroadcasting, as well as making the work available to the public in a manner allowing everyone to have access to it in a place and at time chosen by themselves.

# Transfer of ownership of a copy versus transfer of copyright

- Unless otherwise specified in an agreement, the transfer of ownership of a copy of a work does not transfer the author's economic rights in the work.
- Unless otherwise specified in the contract, the transfer of author's economic rights does not transfer ownership of a copy of the work to the acquirer.
- The purchaser of the original work is obliged to make it available to the author to the extent necessary to exercise the copyright. The purchaser of the original work may, however, demand appropriate security and remuneration for the use from the author.

## Requirement of written form of the contract

- An agreement for the transfer of author's economic rights shall be in writing or else shall be null and void.



# Licensing

- In the absence of an express transfer provision, the creator is deemed to have granted a license.
- A license agreement entitles the licensee to use the work for a period of five years within the territory of the state where the licensee is located, unless otherwise specified in the agreement.
- After the expiration of the period referred to above, the right obtained under the license agreement shall expire.
- The Author may grant authorization to use the work within the fields of exploitation specified in the agreement, specifying the scope, place and time of such exploitation.

## **License for an unlimited period of time**

- If the contract does not provide otherwise, and the license is granted for an indefinite period, the creator may terminate it subject to contractual deadlines, or in their absence one year in advance, at the end of the calendar year.
- A license granted for more than five years shall, upon the expiration of such term, be deemed to have been granted for an indefinite period.

# Computer programs

- Computer programs shall be protected as literary works, except as otherwise provided in this chapter.
- The protection accorded to a computer program extends to all its forms of expression. Ideas and principles which form the basis of any element of a computer program, including the basis of links, shall not be protected.
- The employer owns the economic rights to the computer program created by the employee as a result of performing the duties of the employment relationship, unless the contract provides otherwise.

# **IT Social and professional problems**

## **Lecture**

**Arkadiusz Liber**

# **Criminal and professional liability related to information technology products**

IT Social and professional problems

Arkadiusz Liber

# Lecture - Thematic Sequence

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- **Legal protection of an IT product**
- **Protection of the work as a subject of copyright**
- **Protection of IT devices, networks, data and software under the provisions of the Criminal Code**
- **Forensic examination of an IT product and its environment**

# Responsibility

- **PA-Art. 115.1** Whoever appropriates the authorship or misrepresents the authorship of the whole or part of another's work or artistic performance shall be subject to a fine, the penalty of restriction of liberty or the penalty of deprivation of liberty for up to 3 years.
- (2) Whoever distributes, without indicating the name or pseudonym of the author, another person's work in the original version or as a derivative work, artistic performance, or publicly distorts such work, artistic performance, phonogram, videogram or broadcast shall be subject to the same penalty.
- Whoever infringes another person's copyright or related rights, referred to in Article 16, Article 17, Article 18, Article 19(1), Article 191, Article 86, Article 94(4) or Article 97, or fails to fulfil the obligations set out in Article 193(2), Article 20(1)-(4), Article 40(1) or Article 40(2) in order to obtain a financial benefit, shall be subject to a fine, the penalty of limitation of liberty or deprivation of liberty for a term up to one year.



# Unauthorized Distribution of Another's Work

- **PA-Art. 116** (1) Whoever, without authorization or in violation of the conditions of authorization, distributes another's work in the original version or as a derivative work, artistic performance, phonogram, videogram or broadcast shall be subject to a fine, penalty of restriction of liberty or imprisonment for up to 2 years.
- (2) If the perpetrator commits the act specified in paragraph (1) in order to obtain a financial benefit, he shall be subject to the penalty of deprivation of liberty for up to 3 years.
- (3) If the perpetrator has made the commission of the offence specified in paragraph (1) a regular source of income or organises or directs the criminal activity specified in paragraph (1), he shall be subject to the penalty of deprivation of liberty for a term of between 6 months and 5 years.
- (4) If the perpetrator of the act specified in paragraph (1) acts unintentionally, he shall be subject to a fine, the penalty of restriction of liberty or the penalty of deprivation of liberty for up to one year.

# Unauthorized Reproduction of a Work

- **PA-Art. 117 (1)** Whoever, without authorization or in violation of its terms, for the purpose of distribution fixes or reproduces another's work in the original version or in the form of a development, artistic performance, phonogram, videogram or broadcast shall be subject to a fine, the penalty of limitation of liberty or deprivation of liberty for up to 2 years.
- (2) If the perpetrator has made a regular source of income out of committing the offence specified in paragraph (1), or organises or directs the criminal activity specified in paragraph (1), he shall be subject to the penalty of deprivation of liberty for up to 3 years.

# Unauthorized acquisition and aiding and abetting disposal

- **PA-Art. 118.1** Whoever, in order to gain a material profit, acquires or assists in disposing of an object which is a carrier of a work, artistic performance, phonogram or videogram distributed or reproduced without permission or in violation of its terms, or accepts such an object or assists in concealing it, shall be subject to the penalty of deprivation of liberty for a term of between 3 months and 5 years.
- (2) If the perpetrator has made a regular source of income out of committing the offence specified in paragraph (1), or organises or directs the criminal activity specified in paragraph (1), he shall be subject to the penalty of deprivation of liberty for a term of between 1 and 5 years.
- (3) If, on the basis of the accompanying circumstances, the perpetrator of the offence specified in paragraph (1) or (2) should and may presume that the object has been obtained by means of a prohibited act, he shall be subject to a fine, the penalty of restriction of liberty or the penalty of deprivation of liberty for up to 2 years.

## Removal of technical safeguards

- **PA-Art. 1181 (1)** Whoever manufactures devices or their components intended for the unauthorized removal or circumvention of effective technical protections against the reproduction, re-recording or reproduction of works or objects of related rights, or trades in such devices or their components, or advertises them for sale or lease, shall be subject to a fine, penalty of restriction of liberty or imprisonment for up to 3 years.
- (2) Whoever possesses, stores or uses devices or their components, referred to in section 1, shall be subject to a fine, the penalty of limitation of liberty or deprivation of liberty for up to one year.

# Unauthorized access to information

- **CC-Art. 267.** § 1. Whoever, without authorization, gains access to information not intended for him by opening a closed magazine, by connecting to a telecommunications network or by breaking or bypassing electronic, magnetic, computer or other special protection thereof, shall be subject to a fine, the penalty of limitation of liberty or deprivation of liberty for up to 2 years.
- § 2. The same punishment shall be imposed on anyone who without authorization gains access to all or part of an information system.

## Unauthorized access to information cont.

- § 3. The same punishment shall be imposed on anyone who, in order to obtain information to which he is not entitled, establishes or uses a listening or visual device or other device or software.
- The same punishment shall be imposed on anyone who discloses information obtained in the manner specified in § 1-3 to another person.
- (5) The prosecution of the offence specified in § 1-4 shall take place on the motion of the victim.

# Information Record Destruction

- **CC-Art. 268.** § 1. Whoever, without being authorized to do so, destroys, damages, deletes or alters a record of material information or otherwise prevents or substantially impedes an authorized person from learning about it, shall be subject to a fine, penalty of restriction of liberty or imprisonment for up to 2 years.
- § 2. If the act specified in § 1 concerns a record on a computer data carrier, the perpetrator shall be subject to the penalty of deprivation of liberty for up to 3 years.
- § 3. Whoever, while committing the act specified in § 1 or 2, causes substantial damage to property, shall be subject to the penalty of deprivation of liberty for a term of between 3 months and 5 years.
- (4) The prosecution of the offence specified in § 1-3 shall take place on the motion of the injured person.

## Destroying, damaging, deleting, altering, obstructing access to data

- **KK-Art. 268a**. Whoever, without being authorized to do so, destroys, damages, deletes, alters or impedes access to computer data or substantially interferes with or prevents automatic processing, storage or transfer of such data, shall be subject to the penalty of deprivation of liberty for up to 3 years.
- § 2. Whoever, while committing the act specified in § 1, causes substantial damage to property, shall be subject to the penalty of deprivation of liberty for a term of between 3 months and 5 years.
- (3) The prosecution of the offence specified in § 1 or 2 shall take place on the motion of the injured person.



## **Destruction of sensitive data**

- **KK-Art. 269.** § 1. Whoever destroys, damages, deletes or alters computer data whoever destroys, damages, deletes or alters computer data of special importance for national defense, security of communication, functioning of government administration, other state authority or state institution or local government or interferes with or prevents automatic processing, storage or communication of such data, shall be subject to the penalty of deprivation of liberty for a term between 6 months and 8 years.
- § 2. The same punishment shall be imposed on anyone who commits the act specified in § 1 by destroying or replacing a computer data carrier or destroying or damaging a device used for automatic processing, storage or transmission of computer data.

## Interference with transmission and computer system operation

- **KK-Art. 269a**. Whoever, without being authorized to do so, by transmitting, destroying, deleting, damaging, obstructing access to or altering computer data, substantially interferes with the operation of a computer system or data communications network, shall be subject to the penalty of deprivation of liberty from 3 months up to 5 years.

# Interference with transmission and computer system operation

- **KK-Art. 269b.** § 1. Whoever creates, acquires, disposes of or makes available to other persons devices or computer software adapted for the commission of the offense specified in Article 165 § 1 item 4, Article 267 § 3, Article 268a § 1 or § 2 in connection with § 1, Article 269 § 1 or 2 or Article 269a, as well as computer passwords, access codes or other data enabling unauthorized access to information stored in an information system, data communications system or data communications network, shall be subject to the penalty of deprivation of liberty for a term of between 3 months and 5 years.
- § 1a. Whoever acts with the sole purpose of securing an IT system, an ICT system or an ICT network against committing an offence specified in this provision, or develops a method of such securing, shall not commit the offence specified in § 1.
- (2) If convicted of an offense specified in § 1, the court shall order the forfeiture of the objects specified therein, and may order their forfeiture if they were not the property of the offender.

## Interference with transmission and computer system operation

- **KK-Art. 269c.** Whoever acts solely for the purpose of securing an information system, an ICT system or an ICT network, or developing a method of such securing, and promptly notifies the operator of such system or network of disclosed threats, and his action did not violate public or private interests and did not cause damage, shall not be subject to punishment for the offence specified in Article 267 § 2 or Article 269a.

# Theft of property and theft of a computer program

- **KK-Art. 278.** § 1. Whoever takes another's movable property for the purpose of appropriation shall be subject to the penalty of deprivation of liberty for a term of between 3 months and 5 years.
- § 2. The same punishment shall be imposed on anyone who, without the authorized person's consent, obtains another's computer program for financial gain.
- § 3 In the case of minor gravity, the perpetrator shall be subject to a fine, penalty of restriction of liberty or imprisonment for up to one year.
- § 4. If the theft is committed to the detriment of a person who is closest to the victim, prosecution shall take place at the request of the victim.
- (5) The provisions of § 1, 3 and 4 shall apply mutatis mutandis to the theft of energy or a card authorizing the withdrawal of money from a bank machine.

# **IT Social and professional problems**

## **Lecture**

**Arkadiusz Liber**

# **IT product and industrial property**

IT Social and professional problems

Arkadiusz Liber

# Lecture - Thematic Sequence

|                                        |                                   |                                                                                 |                                   |                                         |
|----------------------------------------|-----------------------------------|---------------------------------------------------------------------------------|-----------------------------------|-----------------------------------------|
| Introduction                           | IT product                        | Civil and intellectual property<br>intellectual property<br>in computer science | PI - as a subject of<br>copyright | PI - as a subject of<br>copyright cont. |
| PI Personal Rights                     | Property Rights<br>Relating to PI | Permitted use of PI<br>and parts thereof                                        | PI in science<br>and education    | Production of PI                        |
| Criminal and<br>professional liability | PI and industrial<br>property     | Patents and<br>trademarks                                                       | Licenses                          | Ethics. Risks. Codes                    |



- **Industrial property**
- **Applicability of patents in IT product protection**
- **Social and legal problems of patenting inventions**

# Industrial Property Law

- Industrial property law originated earlier than copyright law.
- In the early days, it was closely associated with technical equipment and technology.
- The introduction of copyright made it possible to protect objects that are not related to technology or engineering, The objects protected by copyright do not have to have any material value, nor do they have to be useful for anything.
- The Paris Convention of 20.03.1883 is considered the basic document in the field of industrial property law. At present, the Paris Convention is understood as the Stockholm Act, which amends the original Paris Convention.

# Objects of industrial property right

- The objects of industrial property rights are:
- inventions
- utility models,
- industrial designs,
- trademarks,
- geographical indications,
- topography of integrated circuits.

# Inventions

- An invention is considered new if it is not part of the state of the art.
- The state of the art is defined as anything that, prior to the date by which priority to a patent is determined, has been made available to the public by written or oral description, by use, exhibition or disclosure in any other manner.
- An invention is considered to have an inventive step if the invention is not apparent to the inventor from the state of the art.

# Inventions - Conditions for exclusion

- In particular, the following are not considered as inventions:
  - 1) discoveries, scientific theories, and mathematical methods;
  - 2) creations of a purely aesthetic nature;
  - 3) Schemes, rules and methods for carrying out thought processes, playing games or conducting business;
  - <sup>41)</sup> creations or methods of which:
    - (a) the possibility of use cannot be demonstrated or
    - (b) the use will not produce the result expected by the applicant
      - in light of generally accepted and recognized principles of science;
  - 5) computer programs;
  - 6) present information.

# Utility models

- A utility model is a new and industrially applicable technical solution, related to the shape or structure of an object of durable form or to an object composed of functionally connected parts of durable form.
- The requirement of the form of the object shall also be deemed to have been met if the product according to the utility model is defined spatially by repetitive elements with a constant ratio of sizes.
- The requirement for an object to have a permanent form shall also be met if the product according to the utility model changes its form due to due to its intended use.

# Utility model protection

- A utility model can be granted a protection right.
- By obtaining the right of protection one acquires the right to exclusive use of the utility model in a commercial or professional manner throughout the territory of the Republic of Poland.
- The term of the protection right is ten years from the date of filing the utility model application with the Patent Office.
- The subject matter of the right of protection shall be determined by the claims contained in the specification of the utility model.

# Industrial designs 1

- An industrial design is a new and individual character of a product or a part thereof, given in particular by the features of the line, contour, shape, color, texture or material of the product and by its ornamentation.
- A product is any object created in an industrial or handicraft manner, including in particular packaging, graphic symbols and typographic typefaces, but excluding computer programs.



## Industrial designs 2

- A product is also considered to be:
  - 1)An object consisting of a number of interchangeable components that allow it to be disassembled and reassembled (a complex product);
  - (2) a replaceable component part of a composite article if, when incorporated into the composite article, it remains visible during its ordinary use, by which he means any use, excluding maintenance, service, or repair.

## Industrial designs 3

- An industrial design possesses individual character if the overall impression it produces on an informed user differs from the overall impression produced on him by a design made available to the public before the date according to which priority is determined.
- In assessing the individual character of an industrial design, the extent of creative freedom in developing the design is taken into account.

## Industrial designs 4

- An industrial design is granted a right of registration.
- By obtaining the right in registration, the holder acquires the right to exclusive use of the industrial design in a commercial or professional manner in the entire territory of the Republic of Poland.
- The rightholder may prohibit third parties from making, offering, putting on the market, importing, exporting or using a product in which the design is incorporated or to which it is applied, or storing such a product for such purposes.
- The right of registration of an industrial design extends to any design that does not produce a different overall impression on an informed user.

# Trademarks

- A trade mark can be any sign capable of distinguishing the goods of one undertaking from those of another undertaking and capable of being represented in the trade mark register in such a way as to identify unambiguously and precisely the subject matter of the protection granted.
- A trademark can be, in particular, a word, including a surname, a drawing, a letter, a number, a color, a spatial form, including the shape of goods or packaging, as well as a sound.

## Trade marks - supplementary terms

- Trademarks - are also service marks;
- goods - are in particular industrial products, crafts, agricultural produce and natural products, especially water, minerals, raw materials, as well as some services;
- counterfeit trade marks - are illegal identical marks or marks that cannot be distinguished under normal conditions of trade from registered marks, for goods covered by the right of protection;
- Prior marks - are marks applied for or registered with an earlier priority.

# Geographical indications 1

- Geographical indications, as defined by the Act, are verbal signs referring directly or indirectly to the name of a place, locality, region or country (territory) that identify a good as originating from that territory, if a certain quality, good reputation or other characteristics of the good are attributed primarily to the geographical origin of the good.
- Foreign geographical indications may be protected in Poland only if the indication enjoys protection in its country of origin.

## Geographical indications 2

Geographic designations are:

- (1) Regional names as designations used to distinguish goods that:
  - come from a particular area, and
  - have special characteristics that are exclusively or predominantly due to the influence of the geographical environment-which includes both natural and human factors-whose production or processing occurs on the site;
- (2) designations of origin as signs used to distinguish goods:
  - a) originating from a particular area, and
  - b) having certain specific properties or other characteristics attributable to their geographical origin, i.e. the area where they were produced or processed.

## Geographical indications 3

- Geographical indications are also understood as indications used for goods which are produced from raw materials or intermediate products originating from a defined area, larger than the area of production or processing of the goods, if they are prepared under specific conditions and there is a system of control of compliance with those conditions.
- The right of registration for a geographical indication may be cancelled at the request of any person who has a legal interest therein, if the person proves that the statutory conditions required for obtaining the right have not been met.



# Topography of integrated circuits 1

- Topography of an integrated circuit, hereinafter referred to as "topography", means a solution consisting of a spatial arrangement, expressed in any manner, of elements, at least one of which is an active element, and of all or part of the connections of the integrated circuit.
- An integrated circuit is understood to be a single or multilayer spatial product formed by elements of semiconductor material forming a continuous layer, their interconnected conducting and insulating areas inseparably coupled together to perform electronic functions.

## Topography of integrated circuits 2

- Topography is considered original if it is the result of the creator's intellectual work and is not generally known at the time of its creation.
- A topography consisting of common elements is granted registration rights only to the extent that the combination of those elements is original.
- No registration right shall be granted for a topography unequivocally resulting from the function of the integrated circuit in which it is used.

# Rights of the author of the object of industrial property right

The creator of an invention, utility model, industrial design or integrated circuit topography is entitled to

- obtaining a patent, a protection right or a right in registration;
- wages;
- to be named as the author in descriptions, registers, and other documents and publications.

The creator of a rationalization project accepted by the entrepreneur for use shall be entitled to remuneration as specified in the rationalization regulations adopted by the entrepreneur.

Only solutions that are not the subject of industrial property rights can be included in a rationalization project.

# Rights of the author of the object of industrial property right

- Co-authors of an invention, utility model, industrial design or integrated circuit topography shall be entitled jointly.
- If an invention, utility model or industrial design is made as a result of the performance by the inventor of duties resulting from the employment relationship or the right to obtain a patent for the invention shall belong to the employer or contracting party, unless the parties agree otherwise.

# **Rights of the author of the object of industrial property right**

- If an invention, utility model or industrial design is made by an inventor with the help of an entrepreneur, the entrepreneur may use to use this invention, utility model or industrial design himself. In a contract for granting aid, the parties may agree that the entrepreneur is entitled, in whole or in part, to a patent for the invention.

# The right to obtain a patent or a protection right

- The right to obtain a patent for an invention, a utility model or an industrial design is transferable and inheritable.
- The contract for the transfer of this right shall be in writing; otherwise it shall be null and void.
- The priority to obtain a patent, a utility model or an industrial design shall be determined by the date of filing the application for a patent, utility model or industrial design with the Patent Office.

# **Application for registration of an industrial property right with the Patent Office (1)**

- The application shall be deemed to have been filed on the date on which it is received by the Patent Office in writing, also by means of fax or in electronic form.
- If the application is sent by fax, the original application should reach the Patent Office within 30 days from the date of sending the application by fax. This period cannot be restored.
- If the submission sent by fax is illegible or is not the same to the original delivered, the date of notification will be the date on which the original was delivered.

## **Application for registration of an industrial property right with the Patent Office (2)**

- The above provision shall apply mutatis mutandis if the original is received after the deadline referred to above; in this case, the faxed application shall be considered null and void.
- Where an application sent in electronic form contains malware, the Patent Office is not obliged to open such correspondence and further process it. In such a case, as well as if the transmitted application is illegible, the effect of illegibility of the application does not arise.
- The electronic form of the notification shall be understood as the form established at sending the notification by means of a telecommunication network or on a computer data carrier



## **Application for registration of an industrial property right with the Patent Office (3)**

- In the cases referred to above, as long as some of the parts of the application are illegible, they shall be deemed not to have been filed.
- The Patent Office shall immediately notify the applicant, using the same means of communication, that the application sent by fax is wholly or partially illegible or one of the cases has occurred where the electronic mail address or the identity of the applicant and his address can be established and the security of the Patent Office's information and communications system is not compromised and technical considerations allow.

## **Application for registration of an object of industrial property rights exhibited at an exhibition**

- If an invention, utility model or industrial design, which is the subject of a first proper application, has been previously exhibited at an exhibition and enjoyed, from the date of its exhibition until the date of filing the application, provisional protection provided for in the Paris Convention, the priority to obtain a patent, utility model or industrial design, as well as the beginning of the periods prescribed therein for filing an application with the Patent Office shall be determined by the date of exhibition of that invention, utility model or industrial design at the exhibition.

## **Simultaneous filing of an application for an industrial property right**

- If an application for an invention, utility model or industrial design is filed independently by two or more persons who enjoy  
The right to obtain a patent, utility model or industrial design is vested in each of them.

## **Transfer of the object of industrial property right by the creator to the entrepreneur**

- An inventor of an invention, utility model or industrial design who is entitled to a patent, protection right or right in registration may transfer that right, free of charge or for an agreed consideration, to an entrepreneur or may transfer the invention, utility model or industrial design to the entrepreneur for exploitation.

# Remuneration for the author of an industrial property right

- If the parties have not agreed on the amount of remuneration, the remuneration shall be fixed in a fair proportion to the entrepreneur's profits from the invention, utility model or industrial design, taking into consideration the circumstances in which the invention, utility model or industrial design was made, and in particular the extent of the assistance rendered to the author in making the invention, utility model or industrial design, and the extent of the author's obligations in connection with making the invention, utility model or industrial design.
- Unless otherwise specified in the contract, the remuneration shall be paid in whole or in parts.
- The entire remuneration shall be paid at the latest within two months from the date of obtaining the first profits from the invention, utility model or industrial design. If the remuneration is paid in parts, the first part shall be paid within the time limit referred to in the previous sentence, and the remaining parts shall be paid within two months after the end of each year at the latest, but within a period not exceeding 5 years from the date of obtaining the first benefits.

# **Criminal provisions in industrial property law 1**

- Whoever attributes to himself the authorship or misleads another person as to the authorship of another's invention project, or otherwise infringes the rights of the author of the invention project, shall be subject to a fine, the penalty of limitation of liberty or deprivation of liberty for up to one year.
- If the perpetrator commits the act specified above for the purpose of obtaining a financial or personal benefit, he shall be subject to a fine, penalty of restriction of liberty or imprisonment for up to 2 years.

## Criminal provisions in industrial property law 2

- Whoever, without being entitled to a patent, protection right or right in registration, applies for somebody else's invention, utility model, industrial design or topography of an integrated circuit in order to obtain a patent, protection right or right in registration, shall be subject to a fine, penalty of restriction of liberty or deprivation of liberty for up to 2 years.
- The same penalty shall be imposed on anyone who discloses information obtained in respect of another person's invention, utility model, industrial design or topography of an integrated circuit, or otherwise prevents the obtaining of a patent, protection right or right in registration.
- If the perpetrator of the act specified above acts unintentionally, being obliged to keep the obtained information secret, he is subject to a fine.

## **Criminal provisions in the field of industrial property law 3**

- Whoever, with a view to marketing, marks objects not protected by a patent, a supplementary protection right, a utility model protection right or the right of registration of an industrial design, a topography of an integrated circuit or registration of a geographical indication, with inscriptions or drawings aiming at creating a misleading impression that such objects enjoy such protection, shall be subject to a fine or to arrest.
- The same punishment shall be imposed on anyone who, knowing that the objects mentioned above are falsely marked, places such objects on the market or prepares or stores them for marketing purposes, or gives information in announcements, notices or otherwise in a manner that is intended to create the impression that such objects are protected by the law.



## **Criminal provisions in the field of industrial property law 4**

- Whoever places on the market goods bearing a trademark with a distinctive sign intended to create a misleading impression that such items benefit from such protection shall be subject to a fine.
- Prosecution shall be upon the request of the victim.

# **IT Social and professional problems**

## **Lecture**

**Arkadiusz Liber**

# **Patents and trademarks and their use in developing information technology products**

IT Social and professional problems

Arkadiusz Liber

# Lecture - Thematic Sequence

|                                        |                                   |                                                                                 |                                   |                                         |
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| Introduction                           | IT product                        | Civil and intellectual property<br>intellectual property<br>in computer science | PI - as a subject of<br>copyright | PI - as a subject of<br>copyright cont. |
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| Criminal and<br>professional liability | PI and industrial<br>property     | Patents and<br>trademarks                                                       | Licenses                          | Ethics. Risks. Codes                    |



- **Patents**
- **Licensing of an IT product and an industrial product**
- **Types of licenses and the scope of their application**
- **Example software licenses**
- **Risks in the IT profession**

# Patents

- Patents are granted - regardless of the field of technology - for inventions that are new, involve an inventive step and are capable of industrial application.
- A patentee may obtain a patent for an improvement or supplement to an invention that has the features of the invention and cannot be used on its own (a supplementary patent). It is also possible to obtain a supplementary patent to an already obtained supplementary patent.

# Patents

Patents are not granted for:

- inventions, the use of which would be contrary to public order or morality; it shall not be considered contrary to public order to use of an invention merely because it is prohibited by law;
- plant varieties or breeds of animals and purely biological means of plant or animal breeding, as well as products obtained by such means; this provision shall not apply to microbiological or other technical means or to products obtained by such means, unless they are plant varieties or breeds of animals;
- methods of treatment of human beings and animals by surgical or therapeutic methods and methods of diagnosis used on human beings or animals; this provision does not apply to products, and in particular to substances or mixtures used in diagnosis or treatment.

A method of plant or animal breeding is purely biological if it consists entirely of natural phenomena, such as crossing or selection.

# Patents

- By obtaining a patent, one acquires the right to exclusive use of the invention for profit or in a professional manner in the entire territory of the Republic of Poland.
- The scope of a patent is defined by the claims of the patent, which are contained in the patent description. The description of the invention and the drawings can be used to interpret the claims of the patent.
- The term of a patent is 20 years from the date of filing the invention application in the Patent Office.



# Patent and license

The patentee may prohibit a third party, without his consent, from using the invention in a commercial or professional manner consisting of:

- (1) making, using, offering, marketing, storing, or depositing products which are the subject matter of the invention, or exporting or importing them for such purposes, or
- (2) to use the method which is the subject of the invention, and to use, offer, market, store or stock products obtained directly by such method, or to export or import them for such purposes.

The patent owner can contractually authorize (license) another person to use his invention (license agreement).

# Transferability of a patent

- The patent is transferable and inheritable.
- A patent transfer agreement must be made in writing under pain of nullity.
- The transfer of the patent becomes effective against third parties when the transfer is recorded in the patent register.
- The creation of a registered pledge of a patent is effective upon entry in the register of pledges and is subject to disclosure in the patent register.
- The Patent Office, at the request of the competent authority, shall make an entry in the Patent Register of information on the seizure of a patent.

## Prohibition on abuse of patent right

- The holder of a patent or license may not abuse his right in particular by preventing a third party from using the invention when it is necessary to meet the needs of the domestic market and in particular where the public interest so requires and the product is available to the public in insufficient quantity or quality or at excessively high prices.
- It shall not be considered an abuse of the right referred to above to prevent third parties from using the invention during a period of 3 years from the date the patent was granted.
- The Patent Office may request from the patentee or licensee any explanation of the scope of use of the invention in order to determine whether the patent is being abused.

# Patents for secret inventions 1

- An invention made by a Polish citizen may be deemed secret if it concerns the defence or security of the State.
- Inventions relating to the defense of the State are, in particular, types of weapons or military equipment and methods of warfare.
- Inventions relating to State security are, in particular, technical means used by state services authorized to perform operational and exploratory activities, as well as new types of equipment and equipment and methods of their use by these services.

## Patents for secret inventions 2

- A secret invention is an invention that constitutes classified information that has been classified as: "top secret", "secret", "confidential" or "restricted".
- The secrecy of an invention pertaining to State defence or security shall be determined by the Minister of National Defence, the minister in charge of internal affairs or the Head of the Internal Security Agency, respectively.
- An application for a secret invention may be filed with the Patent Office only to claim priority to a patent. During the period in which the claimed invention remains secret, the Patent Office shall not examine the application.

## Patents for secret inventions 3

- The right to obtain a patent for a secret invention applied for in the Patent Office to claim priority shall be transferred, subject to compensation, to the State Treasury represented by the Minister of National Defense, the minister in charge of internal affairs or the Head of the Internal Security Agency, respectively.
- The amount of compensation is determined by the market value of the invention.
- If the parties do not agree on the amount and conditions of the compensation, the compensation, determined by the Minister of National Defence, the minister in charge of internal affairs or the Head of the Internal Security Agency, shall be paid from the State budget on a one-off basis or in parts every year, however, not longer than for 5 years.

# Licence agreements for the use of the invention 1

- A license agreement must be in writing to be valid.
- In a license agreement, the use of the invention may be restricted (limited license). If the license agreement does not limit the use of the invention The licensee is entitled to use the invention to the same extent as the licensor (full license).
- The license shall expire at the latest upon expiration of the patent. The parties may provide for a longer term for provisions other than a license, including, in particular, consideration necessary for the use of the invention.

## Invention licensing agreements 2

- If a license agreement does not provide for exclusive use of the invention  
In case a license agreement does not provide for exclusive use of the invention in a specific manner, the granting of a license to one person does not preclude the granting of a license to other persons, as well as simultaneous use of the invention by the patent owner (non-exclusive license).
- The licensee may grant a sublicense only with the permission of the patent holder; granting a sublicense is not allowed.
- A license is subject, at the request of the interested party, to entry in the patent register. The holder of an exclusive licence entered in the register may, unless otherwise provided for in the licence agreement, pursue a claim for infringement of the patent on an equal footing with the holder of the patent.



## License agreements - transfer of knowledge and experience

- Unless otherwise provided in the license agreement, the licensor is obliged to provide the licensee with all the information and technical experience in his possession at the time of the license agreement that is necessary for the use of the invention.

## Licence agreements - open licence 1

- The patent holder may file a statement with the Patent Office of willingness to grant a license to use his/her invention (open license). Such a statement may not be revoked or amended.
- The information on the filing of the statement referred to above shall be recorded in the Patent Register.
- If a declaration of willingness to grant an open license is filed, the periodic fees for the protection of the invention shall be reduced by half. This provision shall also apply to a one-off protection fee or to the first periodic fee if this reduced fee is received together with the declaration at the latest within the prescribed time limit.

## License agreements - open license 2

- An open license is full and non-exclusive, and the royalty may not exceed 10% of the benefits received by the licensee in each year of use of the invention, net of expenses.
- An open license is obtained by:
  - entering into a license agreement or
  - commencing to exploit the invention without commencing or before the end of negotiations, in which case the licensee shall notify the licensor thereof in writing within one month of commencing to exploit the invention using the invention.
- Unless the agreement provides otherwise, the licensee shall pay the maximum fee within one month after the end of each calendar year in which the licensee used the invention, unless the licensor's statement provides for a lower fee.

# Licence agreements - compulsory licence 1

- The Patent Office may grant permission to use another person's patented invention (compulsory license) when:
  - it is necessary to prevent or eliminate a threat to the security of the State, in particular in the field of defence, public order, protection of human life and health and protection of the natural environment;
  - the patent is found to be abused, or
  - it is found that the owner of a patent with an earlier priority (earlier patent) prevents, by not agreeing to a licensing agreement, the use of the patented invention (dependent patent), the use of which would fall within the scope of the earlier patent, from satisfying the needs of the national market; in this case, the owner of the earlier patent may request permission to use the invention that is the subject of the dependent patent (cross-license).

## Licence agreements - compulsory licence 2

- In order to grant a compulsory license to use an invention, it must be established that the use of the invention that is the subject of the dependent patent, if they are inventions relating to the same subject matter, introduces significant technical progress of considerable economic interest. In the case of an invention relating to semiconductor technology, a compulsory license may be granted only to counteract unreasonably restrictive practices.
- The Patent Office, upon determining that a patent is being abused, may decide that a compulsory license may be sought and shall announce this in the Patent Office News.
- A compulsory license may be granted if the applicant demonstrates that he or she has previously made a good faith effort to obtain the license. Compliance with this condition is not required for the grant of a compulsory license to prevent or remove a threat to state security or in the case of an announcement of the possibility of applying for a compulsory license.

# Patent invalidation

- A patent may be invalidated in whole or in part at the request of any person who shows that:
  - the conditions required to obtain a patent have not been met;
  - the invention has not been presented with sufficient clarity and comprehensiveness to enable an expert to realize the invention;
  - the patent was granted for an invention not covered by the contents of the application or the original application;
  - the patent claims do not clearly and concisely define the subject matter of the claimed protection  
and concise or are not fully supported by the description of the invention.
- The Prosecutor General of the Republic of Poland or the President of the Patent Office may, in the public interest, file a request for patent invalidation or intervene in pending proceedings.

# Patent expiry 1

- The patent expires as a result of:
  - expiration of the period for which it was granted;
  - surrender of the patent by the patentee before the Patent Office, with the consent of the persons who have rights under the patent;
  - failure to pay the periodic fee on time, or
  - permanent loss of use of the invention, due to the lack of biological material needed for the invention, which has become unavailable and cannot be reproduced based on the description.

## Patent expiry 2

- In the cases referred to above, the Patent Office issues decisions declaring the patent to have expired.
- A patent expires on the date on which the event to which the lapse is subject by law has occurred. The date of expiration of the patent should be confirmed in the decision.
- Termination of a patent for failure to pay the periodic fee within the prescribed period shall occur on the date on which the previous term of protection of the invention expired.



# Licenses

IT Social and professional problems

Arkadiusz Liber

# Lecture - Thematic Sequence

|                                        |                                   |                                                                                 |                                   |                                         |
|----------------------------------------|-----------------------------------|---------------------------------------------------------------------------------|-----------------------------------|-----------------------------------------|
| Introduction                           | IT product                        | Civil and intellectual property<br>intellectual property<br>in computer science | PI - as a subject of<br>copyright | PI - as a subject of<br>copyright cont. |
| PI Personal Rights                     | Property Rights<br>Relating to PI | Permitted use of PI<br>and parts thereof                                        | PI in science<br>and education    | Production of PI                        |
| Criminal and<br>professional liability | PI and industrial<br>property     | Patents and<br>trademarks                                                       | Licenses                          | Ethics. Risks. Codes                    |



- **Licensing of an IT product and an industrial product**
- **Types of licenses and the scope of their application**
- **Creative Commons Licenses**
- **Example software licenses**
- **Risks in the IT profession**

# License

- This is permission to use the subject of the license in a specific way.
- From the perspective of copyright law, a license does not necessarily transfer the author's economic right to the licensee.
- The authorizing entity is called the licensor.
- The entity using the license is called the licensee.

# Example software licenses

- Abandonware
- Adware
- AGPL (Affero General Public License)
- Apache License
- APSL (Apple Public Source License)
- Beerware
- BOX
- CDDL (Common Development and Distribution License)
- CPL (Common Public License)
- Donationware
- Freeware
- GNU GPL (GNU General Public License)
- GNU LGPL (GNU Lesser General Public License)
- IDPL (Initial Developer's Public License)
- IPL (InterBase Public License)
- Artistic License
- BSD license
- X11 License (MIT)
- MOLP
- MPL (Mozilla Public License)
- NPL (Netscape Public License)
- OEM (Original Equipment Manufacturer)
- Postcardware
- Public domain (PD)
- Shared Source
- Shareware
- SMSware
- Trial
- WTFPL

# Types of license

- Full license
- Exclusive license
- Non-exclusive license
- Strong license - full + exclusive
- Weak license - the licensor retains the right to use the subject matter of the license.
- Open license
- Cross-license
- Compulsory licence
- Sublicense
- Implied license - the contractor of the work is presumed to grant a license by communicating the results.

# License groups

- Copyleft - A group of licenses that allow a work to be modified and distributed under the same terms as the original work
- © • Non-Copyleft - The group of licenses that do not belong to Copyleft
- Important - Copyleft group includes Creative Common license group with SA attribute without ND and NC attributes

# Construction of a license agreement

- Subject of the license
- Licensee
- Description of the method and conditions of concluding a license agreement
- Description of the nature of the license agreement
- Scope of license
- Licensee's rights
- Licensee's Obligations
- License Term and Termination Regulations
- License fees



## **Problems of Open Source Software Sharing with a Third Party**

- The issue of third party participation. The principles of this participation and the possibility of taking over or excluding copyright.
- The issue of the data carrier on which the software resides. The ability to copy the full Open Source distribution.
- Joint distribution of open source software and software with other licenses
- Open source software in commercial and private devices.

# Creative Commons Licenses

- Creative Commons<sup>1</sup> is an organization that provides a set of licenses of the same name.
- Shared are:
  - License texts (also in some languages other than English),
  - systematized letter and graphic designations,
  - knowledge base and legal simplification.
- This sharing is free. There are links to financial support on the websites of Creative Commons organizations.
- The license texts are prepared for the free use of copyrighted works.
- The Creative Commons Project supports a so-called free culture in which works are treated as common goods.

<sup>1</sup> [www.creativecommons.org](http://www.creativecommons.org)

# Features of the Creative Commons License

- Creative Commons licenses are designed to give more legal freedoms to recipients than fair use.
- Creative Commons licenses are non-exclusive, allowing authors to manage rights to the work in other ways as well.
- Creative Commons licenses contain license modules. These modules can be compiled depending on the needs of the author.
- The icons associated with each licensing module are clear and easy to see. They are well suited for labeling materials.
- Creative Commons licenses are easily recognizable in website metadata allowing for quick categorization of materials.

# Attributes of the Creative Commons License

- There are two groups of CC license attributes permitted operations and mandatory operations.
- The group of permitted operation attributes includes:
  - copying,
  - changing and remixing,
  - dissemination,
  - presenting and performing.
- The mandatory operations attribute group includes:
  - labeling the original author,
  - sharing under the same license,
  - nonprofit use,
  - leaving the work in its original form, without derivative works.

# Labels in Creative Commons licenses

- Creative Commons license terms letter designations:
  - BY - credit to the author,
  - NC - non-commercial use,
  - SA - under the same conditions (this applies to derivative works, which can be created but can only be distributed under a license identical to the original),
  - ND - without dependent works
- Free licenses (letter designations):
  - ZERO - this is a Public Domain license, you can copy, you can change and remix, you can distribute, you can present and perform,
  - BY - this is a zero license with a restriction (mandatory attribute): attribution,
  - BY SA - this is a BY license with a second restriction added: same terms.

# Letter designations for other Creative Commons licenses

- Other Creative Commons licenses (letter designations):
  - BY NC - license derived from the free BY license by adding a restriction: nonprofit use,
  - BY ND - license with restrictions: attribution, no derivative works,
  - BY NC SA - license with restrictions: attribution, non-commercial use, derivative works under the same conditions,
  - BY NC ND - license with restrictions: attribution, non-commercial use, no derivative works.
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- The content change is mainly in the nature of adjusting the license terms to the newer technological and economic situation.

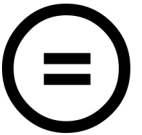
# Creative Commons image markers in Creative Commons license references



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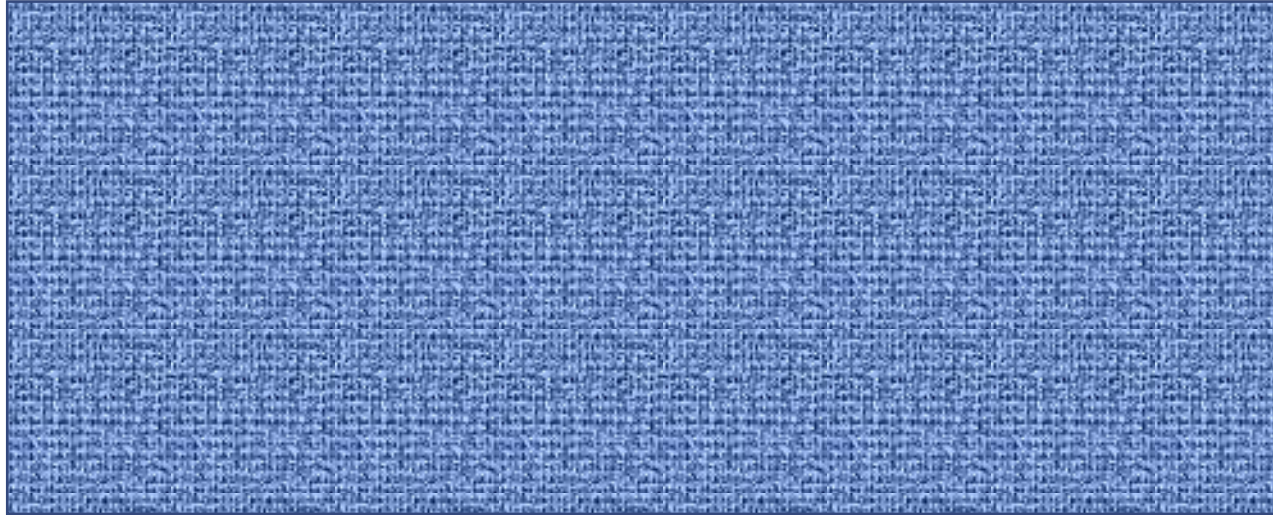
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# **IT Social and professional problems**

## **Lecture**

**Arkadiusz Liber**

# **Ethical issues and risks in the information technology profession**

IT Social and professional problems

Arkadiusz Liber

# Lecture - Thematic Sequence

|                                        |                                   |                                                                                 |                                   |                                         |
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# Issues

- **Ethical issues in the information technology profession**
- **Legal issues**
- **Risks in the IT profession**

# Risks in the IT profession

- Risks in the information technology profession.
- Risk estimation.
- Risk Minimization.
- Professional organizations and collective rights management organizations related to intellectual property.

# Risks

- Risk - the potential possibility of losing an entity that has a certain value.
- Value
  - In ethics, it denotes a certain degree of importance and is sometimes identified with with the concepts of good and evil,
  - in economics, means a measure of the benefit provided by material goods and services,
  - In mathematics, a value can be taken to be one of the elements of a set of values,
- Examples of entities with a certain value: freedom, love, honor, gold, fame, social position, data on a disk

# Risks in the IT profession

- Risks that arise in the IT profession can be divided into:
  - Risks associated with the use of computer equipment,
  - software risk,
  - risk of criminal liability,
  - financial liability risk,
  - risk of losing your license to practice law,
  - image risk,
  - other risks.

# **Estimating the risks associated with Risk assessment for the information technology profession**

- Risk estimation method groups:
  - quantitative methods - are based on determining the effect value and probability of risk occurrence, the effect values are most often determined in monetary values,
  - qualitative methods - based on individual assessment based on good practice and experience,
  - mixed methods - are a combination of quantitative and qualitative methods.



# **Codes of Ethics for the Information Technology Profession**

IT Social and professional problems

Arkadiusz Liber

- **Codes as sets of accepted rules**
- **Mandatory and optional use of codes**
- **Software Engineer's Code of Ethics**
- **PTI Code of Ethics for IT Professionals**

# Codes of ethics and codes of conduct

- Code - a thematically uniform set of rules.
- Codes:
  - Criminal Code,
  - Code of Criminal Procedure,
  - Commercial Code,
  - Family and Guardianship Code,
  - Code of Medical Ethics,
  - Bar Code of Ethics,
  - ...
  - IT Code of Ethics.

# Codes of Information Technology Ethics

- Codes of ethics set standards of **conduct** and **education** in a specific profession. Codes of ethics define professional and ethical responsibilities. Codes also **serve a public information function**.
- There is no uniform code of IT ethics recognized by all communities.
- One of the most widely used is the IEEE-CS/ACM Code of **Software Engineering Ethics**<sup>1</sup>.
- In Poland, it is worth noting the Professional Code of Informatics<sup>2</sup> approved on 29.05.2011 by the Polish Information Technology Association.

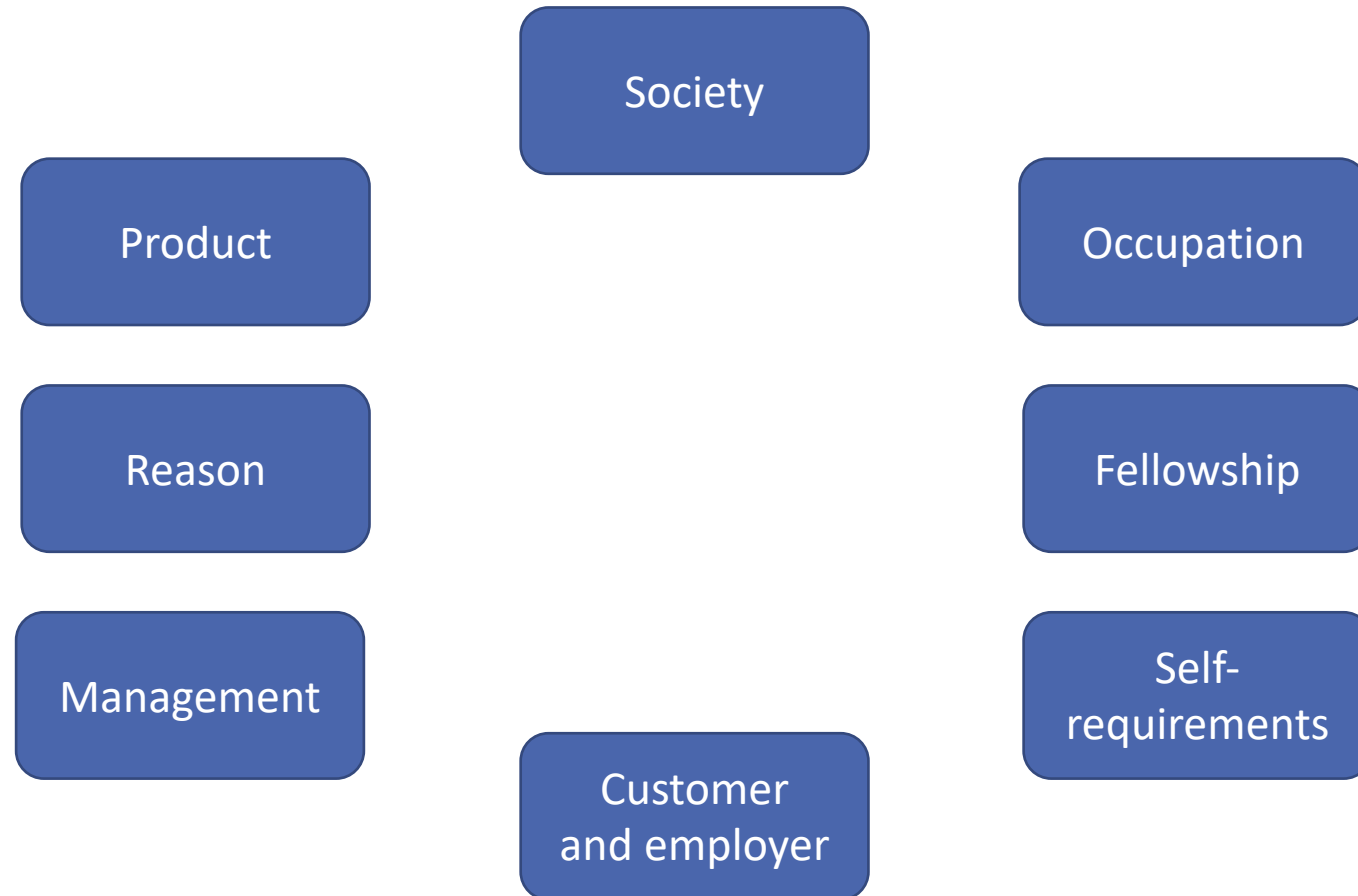
<sup>1</sup>[www.acm.org](http://www.acm.org)

<sup>2</sup>[pti.org.pl](http://pti.org.pl)

# Software Engineer by IEEE-CS/ACM

- A software engineer is a person who directly participates or teaches  
In the field of:
  - software analysis,
  - software specifications,
  - software design,
  - software development,
  - implementation,
  - software maintenance,
  - software testing.

# Components of a software engineer's code of ethics



# Basic principles

- 1 Society.  
Software engineers should act in accordance with the public interest.
- 2 Customer and Employer.  
Software engineers should act in a manner that ensures the interests of their customers and employers.

## Basic principles

- 3 Product.  
Software engineers should ensure the highest possible standards of performance for their products and their modifications.
- 4 Reasoning.  
Software engineers should maintain consistency and independence of their professional judgments.



## Basic principles

- 5 Management.  
Software engineers in managerial and leadership roles support and promote an ethical approach to software development and maintenance.
- 6 Profession.  
Software engineers should further the integration and reputation of the profession consistent with the public interest.

## Basic principles

- 7 - Collegiality.  
Software engineers should behave fairly and be helpful to their colleagues.
- 8 Self-requirements.  
Software engineers should participate throughout their lives in developing their practical professional skills and promote an ethical approach to professional practice.

## Society. Specific rules

- Accepting full responsibility for your work.
- Balancing the interests of the software engineer, employer, customer, and users with respect for the public good.

## Society. Specific rules

- Acceptance of software only if there is a reasonable belief that it is: safe, conforms to specifications, has undergone appropriate testing, does not impair the quality of life, does not reduce privacy, and does not harm the environment. The highest effect of work should be the public good.

## Society. Specific rules

- Disclose to appropriate persons and supervisors the actual or potential danger to the user, the public, the environment that may be associated with the software or related documentation.

## Society. Specific rules

- Collaborate on social **impact of software**, installation, maintenance, support and documentation.
- Be honest and prevent fraud in all forms, especially social fraud, related to related to software, its documentation, methods and tools.

## Society. Specific rules

- Consider issues of physical impairments, resource deployment, economic disadvantages, and other factors that may limit access to software benefits.
- Volunteer to use professional skills for the common good and participate in education related to the discipline represented.

## Client and Employer. Specific rules

- Serve within the scope of his/her **authority** by being **honest and direct** about any limitations on his/her experience and education.
- Do not knowingly use software that has been downloaded or stored in memory **illegally** or **by unknown means**.



## Client and Employer. Specific rules

- Use client or employer property only in an authorized manner with the knowledge and permission of the client or employer.
- Ensure that each document on which the work depends is approved, if required, by a person authorized to approve.

## Client and Employer. Specific rules

- Keep confidential any confidential information acquired in professional work when confidentiality is consistent with the public interest and applicable law.
- Present, document, collect evidence, and immediately report to the client or employer on overcharges, copyright infringement and other problems when they believe a project is about to fail.

## Client and Employer. Specific rules

- Present, document and report to the client or employer significant social issues from which the software or its documentation deviates.
- Accept **not undertaking outside work that is detrimental** to the work performed for the primary employer.
- Promote the **absence of conflicts of interest** with your employer or client as long as a higher principle of ethics is not violated; if so, inform your employer or client of the occurrence of the ethics violation.

# **IT Social and professional problems**

## **Lecture**

**Arkadiusz Liber**

## Product. Specific rules

- Strive for high quality, acceptable cost, and a reasonable schedule in a manner that is clear and acceptable to the employer and customer available for review by the customer and the public.
- Provide appropriate and achievable goals and measures for each ongoing or proposed project.

## Product. Specific rules

- Identify, define and apply ethical, economic, cultural, legal and environmental principles to the project being performed.
- Ensure that the person performing or proposing the project has qualifications that are a combination of education, practice, and experience.

## Product. Specific rules

- Ensure that appropriate methods are used for the performed and proposed projects.
- Work with available standards, deviating from them only when technically or ethically justified.

## Product. Specific rules

- Provide fully understandable specifications for the program being implemented.
- Ensure that the specification of the implemented program is well documented, is consistent with the user requirements specification and has appropriate approvals.



## Product. Specific rules

- Provide realistic estimates of costs, timelines, human resources, and expenses for ongoing and proposed projects with the inaccuracies of those estimates.
- Ensure adequate testing, debugging, and review of executed software and documentation.

## Product. Specific rules

- Solutions and adaptations of significant problems related to the performed project.
- Work on software development and documentation while respecting the privacy of those who come into contact with it.

## Product. Specific rules

- You should be careful to use only accurate, legally verified and ethically verified data in a properly authorized manner.
- Maintain data integrity and be wary of overage and ideal data.
- Maintain the software with the same professionalism as when it was created.

## Reasoning. Specific rules

- Be technically sound with preserving and maintaining human values.
- You should only approve documents prepared under your supervision or in your area of expertise.

## Reasoning. Specific rules

- Professional objectivity must be maintained due to the software and associated documentation.
- Do not accept deceptive financial practices such as bribery, double billing and other improper financial practices.

## Product. Specific rules

- Exclude yourself from all parties with a conflict of interest if it is not possible to truly bypass or remove them.
- Refuse to participate as a member or advisor in private or government software-related organizations with which the employer or client may have a conflict of interest.

## Management. Specific rules

- Ensure good project management with effective procedures related to quality promotion and risk reduction.
- Make software engineers aware of the standards before implementing them.

## Management. Specific rules

- Ensure that software engineers are familiar with policies and procedures for securing passwords, files, and confidential information for the employer and others.
- Assign work only upon proof of appropriate education and experience.



## Management. Specific rules

- Provide realistic quantitative estimates of costs, plans, personnel, quality, and expenditures of each ongoing and planned project along with along with providing the accuracy of those estimates.
- Engage potential software engineers only by fully and accurate description of the terms and conditions of employment.

## Management. Specific rules

- Offer fair and appropriate compensation.
- Do not use undeserved preemption in obtaining a position for which the person is sufficiently qualified.

## Management. Specific rules

- Ensure a fair contract for ownership of software, processes, research, records, or other copyrights associated with the software engineer's contributions.
- Observations should be made regarding violations of employer policies or provisions of this code.

## Management. Specific rules

- Do not request anything from the software engineer that does not comply with this code.
- No one should be penalized for expressing ethical concerns about a project.

## Occupation. Specific rules

- Organizational activities in support of ethical conduct should be developed.
- The public's knowledge of software engineering should be promoted.

## Occupation. Specific rules

- You should expand your knowledge of software engineering by participating in professional organizations, meetings, and publications.
- As a member of the software engineering profession, you should support others to act in accordance with this Code.

## Occupation. Specific rules

- Do not promote your own interests at the expense of your professional community, client or employer.
- Comply with all laws governing the performance of the work except those not in the public interest.

## Occupation. Specific rules

- Be accurate in determining the characteristics of the software being executed, avoiding not only false clauses but also speculative, thoughtless, deceptive, misleading, or questionable clauses.
- Take responsibility for detecting, correcting and reporting errors in software and accompanying documentation.



# Occupation

- Ensure that customers, employers, and reviewers know that the software engineer adheres to this code of ethics and the consequences of adhering to it.
- Avoid business and organizational relationships that conflict with with this code.
- It should be understood that violation of this code is incompatible with the practice of software engineering.
- It should be made clear to those interested in violating this code that such violations are detectable, counterproductive and dangerous.

## Occupation. Specific rules

- Report significant violations of the Code of Ethics to appropriate representatives if it is impossible, unsafe or counterproductive to consult with violators.

## Collegiality. Specific Rules

- Colleagues should be encouraged to follow this code.
- Accompany colleagues in their professional development.

## Collegiality. Specific Rules

- Trust the work of others and refrain from not giving that trust.
- Review the work of others in an objective, honest, and well-documented manner.

## Occupation. Specific rules

- It is important to honestly listen to the opinions, concerns and complaints of colleagues.
- Accompany colleagues in maintaining current work standards including policies and procedures for protecting passwords, files, confidential information, and security in its entirety.

## Occupation. Specific rules

- One should not foully intervene in the careers of colleagues, however one should consider employer, customer, and societal interests that may compel a software engineer to question in good faith the competence of colleagues.
- When situations arise outside of your own area of expertise, consult with other experts in the missing subject matter.

## Requirements of the self. **Specific rules**

- You should take care of your knowledge in the areas of analysis, specification, design, maintenance, testing of software and its documentation together with with management and process execution.
- You should develop your capability to develop secure, reliable, and usable software at an acceptable cost and execution time.

## Requirements of the self. Specific rules

- You should develop your capacity to produce accurate, rich information-rich, well-written documentation.
- Develop your understanding of software and related documentation and the environment in which they are used.



## Requirements of the self. **Specific rules**

- Develop your knowledge of applicable standards and regulations related to software and documentation.
- You should develop your knowledge of the provisions of this code and their interpretation and application to your work.

## Requirements of the self. **Specific rules**

- Do not give unreliable solutions to anyone because of bias.
- Do not be influenced by others in challenging the provisions of this code.
- Personal violations of this code should be recognized as inconsistent with the practice of software engineering.

# PTI Professional Code of Informatics

- Document enacted on 29/05/2011. <sup>1</sup>

# Introductory Part of the PTI KZI Code

- The Polish Information Processing Society, according to its statute, cares about high **ethical** and **professional** level of its members and creates conditions for its improvement.
- Realizing this goal, the Board of the Polish Information Processing Society (PTI), after community consultations, publishes this code of conduct appropriate in all spheres of IT activity and recommends its **application by professionals**, both **members of PTI** and **those who sympathize** with the values professed by them.

- The Code is intended for those **who wish to consider** it applicable to their **computing professional activities** of any profile: scientific, teaching, business and social.
- It is intended to support these individuals in **deciding** in their conscience whether a given **professional** conduct is or is not appropriate, and, if necessary, to assist them in justifying their actions. This role has to be fulfilled not only with respect to **individuals** but also in the relationships of IT professionals **with clients, business partners, colleagues, peers, supervisors, employers and the authorities**.

# Recommendations for appropriate behavior in computing

- 1) Computer scientists, applying computer science as servant knowledge to the fields of its various applications, support the development of these fields without hindering it.
- (2) The application of computer science tools and algorithms is not an end for computer scientists, but a means to solve with respect to the principles of logic, human rights, environment, ergonomics, economics, linguistic correctness, quality standards, and specificity of particular fields of computer science problems presented.
- 3. IT professionals are constantly improving their knowledge, while always presenting their competencies and professional experience according to the facts.

- 4. computer scientists exemplarily respect intellectual property and the rights to protect it.
- (5) IT professionals shall respect the property rights to information and knowledge contained in their employer's and client's information systems.
- (6) IT shall not undertake unauthorized damage to the integrity of the information systems of any entity.

- When presenting specific solutions, freelance computer scientists always try to present the broadest possible spectrum of solutions with analogous functionality and intended use, if any. Similarly, in such activities they try to separate **their own views** on a particular issue **from other** existing views.
- 8. computer scientists engaged in scientific or research and development activities always clearly separate **certain** and already proven **knowledge from the assumptions they make**.
- (9) IT professionals shall always provide their clients with **complete and accurate information** about the **cost** projections and the likely **duration of** the project they are analyzing or a project already in the implementation phase.



- 10. IT provides full and accurate information about the future technical and financial implications of the project.
- (11) IT professionals shall not undertake work for several principals at the same time if their interests might conflict.
- (12) IT professionals avoid playing mutually oppositional roles in projects, such as principal and contractor, subcontractor and controller, programmer and tester.

# Literature

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